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A theory of multilevel interactive equilibrium in neuroAI

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ABSTRACT

We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in NeuroAI and can be viewed as an extension of game theory with relaxed assumptions such as various degrees of uncertainty. At its core, Multilevel Interactive Equilibrium (MIE) generalizes the classical Nash equilibrium to intelligent systems with internal computation. Rather than being defined solely at the level of observable behavior, equilibrium emerges when neural learning dynamics, cognitive representations, and behavioral strategies mutually stabilize between interacting agents. This framework applies uniformly to interactions between two biological brains, two artificial agents, or hybrid human-AI systems. We discuss the emerging applications of multilevel game theory for human-autonomous vehicle driving, human-machine interactions, human-large language model (LLM) interactions, and computational psychiatry. We use simple computer simulations to provide proof-of-concept validation of MIE. We also outline experimental strategies and computational methods to estimate MIE and discuss challenges and prospects in future research.

1. Introduction

For much of its history, the science of intelligence—both biological and artificial—has been framed around a deceptively simple question: how does an agent optimize its behavior in a complex environment? Classical reinforcement learning (RL) and decision theory have largely approached this problem from a single-agent perspective, treating the environment as passive or stationary. But the ecological reality in which the brain evolved is fundamentally different. Organisms rarely act alone. Many real-world behaviors—from economic markets and social cooperation to conflict and negotiation—depend on the ability to anticipate and respond to other agents. In both cooperative and competitive environments, organisms operate within social and adversarial ecosystems, where outcomes depend not only on the environment but also on the actions, intentions, and strategies of other agents. In such settings, behavior becomes intrinsically interactive, and game theory has offered a normative framework for reasoning about strategic interactions in economics, political science, evolutionary biology (von Neumann & Morgenstern, 1944), and computational psychiatry (Montague et al., 2012). Rapid advances in modern artificial intelligence (AI), large language models (LLMs), and robotics present an increasing number of human-AI interactions and NeuroAI scenarios, where the dynamics between interacting agents are complex and the reasoning is conducted under uncertainties (Oulasvirta et al., 2018). Despite some efforts (Ismail, 2024; Razakatinana et al., 2020), to our best knowledge, a unifying theory of characterizing NeuroAI applications across behavioral, cognitive and neural levels is still lacking.

At the core concept of game theory, the Nash equilibrium describes a configuration of strategies in which no agent can unilaterally improve its payoff. The classical equilibrium theory assumes agents with unbounded rationality, complete knowledge of the game structure, and instantaneous access to optimal strategies (Holt & Roth, 2026; Zhang et al., 2014). However, biological brains do not satisfy any of these assumptions. Neural systems are constrained by finite computation, partial observability, noisy information, and limited memory, and must learn strategies gradually through experience, continually adapting to

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the behaviors of other agents who are themselves learning and adapting. Recently, a notion of *neuronic Nash equilibrium* has been proposed in the EEG-derived neural representation space (Zhu, 2025a), generalizing the classic game-theoretic framework for brain-machine interface (BMI)-enabled multi-agent behaviors and extending the notion of inter-individual neural synchronization (Kelso et al., 2013). Additionally, game-theoretic equilbirium convergence has been reported in human-machine interactions, generating insight into human-in-the-loop optimization paradigm for assistive devices (Chasnov et al., 2025).

Decision neuroscience or neuroeconomics investigate how the brain supports strategic interaction (Camerer, 2003, 2009; Glimcher and Fehr, 2014), leading to Bayesian Theory of Mind (ToM) (Yoshida et al., 2008, 2010). Experimental paradigms derived from economic games have revealed neural circuits involved in reasoning about others' actions and intentions. In competitive and cooperative environments, the brain encodes variables relevant to social decision-making, including expected rewards, prediction errors, and beliefs about opponents, suggesting that the brain represents not only the value of actions but also the anticipated strategies of other agents, where successful behavior requires predicting how others will respond. Parallel developments have also taken place in AI. Multi-agent RL (MARL) theory and algorithms propose that learning may occur through repeated interaction between agents that continually adapt to each other (Li & Zhao, 2022; Lowe et al., 2017; Shoham et al., 2003; Su et al., 2026), achieving remarkable successes in domains such as games and robotics (Silver et al., 2017). Therefore, the study of intelligence—whether biological or artificial—confronts the problem of learning and acting within strategic environments populated by other adaptive agents.

Despite these technological advances, a theoretical gap remains. First, classical game theory treats equilibrium purely at the level of observable strategies and payoffs. However, in biological systems, the strategies that appear at the behavioral level are generated by deeper computational processes within the brain. Equilibrium should not be viewed as a single-layer concept but as a multilevel phenomenon emerging from the interaction of neural, cognitive, and behavioral processes. Second, traditional game theory and equilibrium assume that social interactions occur between two human players, yet increasingly popular human-AI interactions pose an emerging need for a new theoretical framework.

To address the gap, here we present a hierarchical multilevel equilibrium framework in which strategic stability emerges from interactions both within-levels and cross-levels of the system. Within this framework, we envision that systems of intelligence, both biological and artificial, can achieve equilibrium at three levels: behavioral, cognitive, and neural levels, forming a conceptual pyramid (Fig. 1). The goal of this theory-favored paper is to outline a proposal and conceptual framework to study emerging problems across various NeuroAI research domains, yet the detailed study of specific applications will be treated elsewhere.

2. Multilevel equilibrium framework

In the hierarchy, at the base lies the *neural level*, where distributed populations of neurons encode value signals, prediction errors, and internal models of other agents. Neural activity implements learning rules—such as RL, active inference (Parr et al., 2022), predictive coding, or synaptic plasticity—that update beliefs and policies during interaction. Above this sits the *cognitive level*, which includes internal representations such as beliefs, intentions, expectations, and models of opponents. At this level, agents maintain structured hypotheses about other agents' strategies and update them through inference. Finally, at the top of the pyramid lies the *behavioral level*, where actions, choices, and policies are expressed and observed. It is at this level that classical game-theoretic concepts: strategies, payoffs, and equilibria, are typically defined.

Within this framework, equilibrium can emerge in several different forms. First, within-level equilibria occur when stability is reached inside a single layer of the system. For example, at the behavioral level, two agents may converge to a stable pattern of strategies resembling a Nash equilibrium in a repeated game. At the cognitive level, equilibrium may correspond to a stable configuration of beliefs about the opponent's strategy. At the neural level, equilibrium may appear as stable attractor dynamics in neural populations within the brain that represent expected value or opponent models, or in the artificial neurons within the artificial neural networks that represent the same information.

Second, equilibria may occur across levels of organization. In biological systems, stable behavior often requires alignment between neural dynamics, cognitive representations, and overt strategies. For example, a behavioral equilibrium in a repeated game may arise only when neural learning signals stabilize the internal beliefs that support the strategy. In this sense, equilibrium can be viewed as a cross-level consistency condition: neural activity generates cognitive models that support behavioral strategies that in turn stabilize neural prediction errors.

More importantly, this hierarchical framework also allows equilibrium to be defined across different types of interacting agents. Traditional game theory primarily considers social or economic interactions among human decision-makers, but developments in AI and NeuroAI have expanded the scope considerably. Not only equilibrium may occur between two AI agents engaging in self-play or competitive learning environments, but it can also emerge between a biological brain and an AI agent, such as in human-AI collaboration, or BMIs with artificial copilots (Lee et al., 2025). In such hybrid systems, stability depends on whether both agents can successfully model and adapt to each other's behavior across multiple levels of representation.

From this perspective, equilibrium is no longer purely a mathematical property of strategic games. Instead, it becomes an emergent property of interacting intelligent systems, arising from coupled multi-agent dynamics across neural, cognitive, and behavioral layers. The pyramid

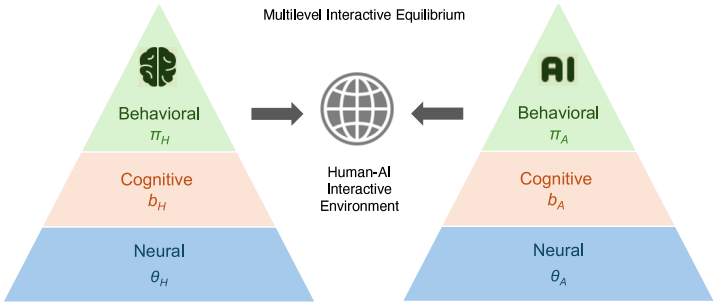


Fig. 1. Conceptual double-pyramid representation of Multilevel Interactive Equilibrium (MIE) during dyadic interactions. Human and AI agents are represented as mirrored hierarchical systems consisting of neural, cognitive, and behavioral layers. Interaction occurs primarily at the behavioral level through the shared environment, while cognitive belief models and neural learning processes adapt through feedback across agents. Stable interaction emerges when neural learning, cognitive inference (or prediction), and behavioral policies mutually stabilize across both pyramids, forming an MIE.

representation helps clarify how these levels interact: neural mechanisms support cognitive models, cognitive models guide behavior, and behavioral outcomes are fed back to update neural representations. When these processes mutually stabilize—either within a single agent or between interacting agents—an equilibrium state may emerge. In this paper, we will use both theoretical construction and computer simulations to support these arguments.

3. A theoretical proposal

Here we propose a constructive theory of multilevel equilibrium of interactive intelligence, where equilibrium is not merely a property of observable strategies but an emergent condition arising from three-timescale, coupled dynamics across neural, cognitive, and behavioral layers of interacting agents. The equilibrium is a fixed point or invariant set of that coupled dynamical system, from which we derive the *within-level equilibrium* (fixed point in one subsystem), *cross-level equilibrium* (simultaneous consistency across subsystems), and *cross-agent equilibrium* (coupling between agent's states). This conceptual framing aligns naturally with NeuroAI because both biological and artificial systems learn, predict, and co-adapt through hierarchical representations.

3.1. Mathematical setting

Let there be N agents $i \in \{1, \cdots, N\}$. At each time $t = 0, 1, 2, \cdots$, we define

- Environment state: $s_t \in \mathcal{S}$;
- Action of agent i: $a_t^i \in \mathcal{A}_i$;
- Joint action: $\boldsymbol{a}_t = (a_t^1, \cdots, a_t^N) \in \mathcal{A} = \prod_i \mathcal{A}_i$;
- Reward (or utility) to agent i: $r_i(s_t, \boldsymbol{a}_t) \in \mathbb{R}$.
- State transition: $s_{t+1} \sim P(\cdot|s_t, \boldsymbol{a}_t)$.

This is a Markov game (a.k.a. stochastic game) (Littman, 1994).

3.2. The three-level pyramid as a state space

To formalize the multilevel structure of intelligent agents, we represent each agent as a hierarchical system consisting of neural, cognitive, and behavioral states. These layers form a conceptual pyramid in which neural mechanisms support cognitive representations, which in turn generate behavioral strategies.

Neural level (substrate of learning). At the base of the pyramid lies the neural or algorithmic substrate that implements learning and inference. For agent i, we denote the neural state by $\theta_t^i \in \Theta_i$, where θ_t^i represents neural parameters or internal weights that govern learning dynamics (e.g., synaptic strengths in biological brains or parameters in artificial neural networks).

Cognitive level (beliefs and internal models). The intermediate layer represents internal cognitive states that encode beliefs about the environment and other agents. We denote the cognitive state by $b_t^i \in \mathcal{B}_i$, where b_t^i represents probabilistic beliefs, latent representations of opponent strategies, or internal models of the environment.

Behavioral level (policies or strategies). At the top of the pyramid lies observable behaviors. Agent i selects actions according to a policy $\pi_t^i \in \Pi_i$, where π_t^i maps environmental states s_t and internal beliefs b_t^i to action distributions.

Multilevel agent state. The complete internal state of agent i is given by the tuple $\boldsymbol{x}_t^i = (\theta_t^i, b_t^i, \pi_t^i)$, which belongs to the product space

$$\mathcal{X}_i = \Theta_i \times \mathcal{B}_i \times \Pi_i.$$

For a system with N interacting agents, the joint multilevel state is

$$\boldsymbol{x}_t = (\boldsymbol{x}_t^1, \boldsymbol{x}_t^2, \ldots, \boldsymbol{x}_t^N) \in \mathcal{X},$$

where $\mathcal{X} = \prod_{i=1}^N \mathcal{X}_i$. This representation defines a hierarchical state space in which neural learning, cognitive inference, and behavioral strategies co-evolve during interaction.

3.3. Coupled dynamics across levels

The multilevel state representation allows us to formalize how neural learning, cognitive inference, and behavioral strategies evolve during multi-agent interaction. These processes form a set of coupled dynamical systems that operate both within each agent and across interacting agents.

Action generation. At time t, agent i produces an action according to its behavioral policy. Conditioned on the environmental state s_t and cognitive belief state b_t^i,

$$a_t^i \sim \pi_t^i(\cdot \mid s_t, b_t^i).$$

The joint action of all agents is $\boldsymbol{a}_t = (a_t^1, a_t^2, \ldots, a_t^N)$. The environment then evolves according to the state transition dynamics $s_{t+1} \sim P(\cdot \mid s_t, a_t)$.

Cognitive update. Agents update their internal models of the environment and other agents based on observations. Let o_t^i denote the observation available to agent i at time t; we assume that belief updates follow an operator

$$b_{t+1}^i = F_i(b_t^i, o_t^i),$$

where F_i may represent Bayesian belief updating, inference over latent opponent strategies, or neural network-based intent estimation.

In Bayesian form, belief update can be written as (Khalvati et al., 2019)

$$b_{t+1}^i(\xi) \propto b_t^i(\xi)\, p(o_t^i \mid \xi),$$

where ξ denotes a latent variable such as an opponent type or behavioral policy.

Although Bayesian updating in the cognitive operator F_i is flexible and fully rational, the MIE framework can be more generic and relax the perfect-rationality assumption. Specifically, F_i can be generalized to allow for reference-dependent valuation, asymmetric sensitivity to gains and losses, and nonlinear belief weighting. For instance, outcomes are evaluated relative to a reference point r, with an asymmetric value function

$$v(\xi) = \begin{cases} (\xi - r)^\alpha, & \xi \geq r, \\ -\lambda(r - \xi)^\beta, & \xi < r, \end{cases}$$

where $\alpha, \beta \in (0, 1]$ capture diminishing sensitivity and $\lambda > 1$ captures loss aversion. Probabilities or beliefs are transformed via a weighting function $w(p)$, such as

$$w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}},$$

with $\gamma \in (0, 1]$ inducing overweighting of small probabilities and underweighting of large ones. Under these extensions, the cognitive operator F_i reflects a distorted aggregation of perceived outcomes and probabilities. A concrete demonstration of reference-dependent preference and probability distortion embedded within multi-agent negotiation and adaptive strategy evolution is shown in Cheng et al. (2025).

Neural learning. At the neural level, internal parameters adapt through learning signals such as prediction errors or gradient updates. Let δ_t^i denote a learning signal available to agent i. Neural parameters evolve as

$$\theta_{t+1}^i = G_i(\theta_t^i, \delta_t^i).$$

A common form is stochastic gradient learning

$$\theta_{t+1}^i = \theta_t^i + \alpha_t \nabla_{\theta^i} J_i(\theta_t^i; \boldsymbol{x}_t^{-i}),$$

where α_t is a learning rate, J_i is the expected return objective for agent i, and $\boldsymbol{x}_t^{-i}$ denotes the joint multilevel state excluding agent i.

Alternatively, parameters can be updated within the MARL framework, including the model-free Actor-Critic model and policy gradient.

Policy adaptation. Behavioral policies are generated from neural parameters and cognitive belief states. We assume that policy updates are described by

$$\pi_{t+1}^i = H_i(\pi_t^i, \theta_{t+1}^i, b_{t+1}^i).$$

For example, a softmax policy may be defined as

$$\pi_{t+1}^i(a \mid s, b) \propto \exp\Big(\beta Q_{\theta_{t+1}^i}(s, a; b)\Big),$$

where Q_θ denotes a value function parameterized by neural parameters θ, and $\beta > 0$ controls the degree of exploration versus exploitation and is often called exploration-exploitation parameter.

Global dynamical system. Combining the environmental state and the multilevel agent states yields a global system state

$$\boldsymbol{z}_t = (s_t, \boldsymbol{x}_t) \in \mathcal{Z},$$

where $\mathcal{Z} = \mathcal{S} \times \mathcal{X}$.

The evolution of the full system can be described by a Markov process

$$\boldsymbol{z}_{t+1} \sim \mathcal{T}(\cdot \mid \boldsymbol{z}_t),$$

where $\mathcal{T}$ is the transition kernel induced jointly by the environmental dynamics and the multilevel update operators $\{F_i, G_i, H_i\}_{i=1}^N$.

Collectively, the above formulation makes explicit that neural learning, cognitive belief updates, and behavioral policies evolve through a coupled dynamical system across interacting agents.

3.4. Multilevel interactive equilibrium

The multilevel dynamical system defined above allows us to formalize the notion of equilibrium in interacting intelligent systems. Classical game theory defines equilibrium purely at the level of behavioral strategies. In contrast, our framework defines the equilibrium across neural learning, cognitive inference, and behavioral policy dynamics.

3.4.1. Preliminaries

Definition 1 (Neural equilibrium)**.** A neural parameter profile $\theta^* = (\theta_1^*, \dots, \theta_N^*)$ is said to be a neural equilibrium if the expected neural learning dynamics are stationary. Formally, for each agent i,

$$\mathbb{E}\big[G_i(\theta_i^*, \delta^i)\big] = \theta_i^*.$$

For gradient-based learning rules, this condition is equivalent to

$$\nabla_{\theta_i} J_i(\theta_i^*) = 0,$$

where J_i denotes the expected return for agent i.

Essentially, neural equilibria define fixed points in the neural state space.

Definition 2 (Cognitive equilibrium)**.** A belief profile $b^* = (b_1^*, \dots, b_N^*)$ is a cognitive equilibrium if beliefs are invariant under the belief update operator induced by interactions. That is, for each agent i,

$$b_i^* = \mathbb{E}\big[F_i(b_i^*, o^i)\big],$$

where the expectation is taken with respect to the stationary distribution of observations generated by the system.

In the two-agent case, cognitive equilibria often occurs if the beliefs of the others are mutually stable.

Definition 3 (Behavioral equilibrium)**.** Given neural and cognitive states (θ, b), a policy profile $\pi^* = (\pi_1^*, \dots, \pi_N^*)$ is a behavioral equilibrium if no agent can improve its expected return through unilateral deviation. Formally,

$$\pi_i^* \in \arg\max_{\pi_i \in \Pi_i} V_i(\pi_i, \pi_{-i}^*; \theta, b),$$

for every agent i, where V_i denotes the expected value of agent i under the joint policy profile.

A special case of two-agent behavioral Nash equilibrium has been discussed earlier (Zhu, 2025b).

Definition 4 (Multilevel Interactive Equilibrium)**.** A tuple (θ^*, b^*, π^*) is a *Multilevel Interactive Equilibrium (MIE)* if all three conditions hold simultaneously:

i. θ^* is a neural equilibrium ("neural stationarity");
ii. b^* is a cognitive equilibrium ("cognitive self-consistency");
iii. π^* is a behavioral equilibrium given (θ^*, b^*) ("behavioral best response").

Equivalently, the multilevel state is a fixed point of the coupled update operators that govern neural learning, cognitive inference, and behavioral adaptation.

The notion of MIE assumes three updates simultaneously reach equilibria; when the coupled systems are only partially unobserved (e.g., neural activity of the human brain is unobservable), we refer to the system that satisfies two of three conditions (e.g., ii + iii) as a marginal MIE. However, the notion of marginal MIE is intended to acknowledge practical limitations in obtaining full-level observability, not to weaken falsifiability (see Section 7.2 for detailed discussions).

3.4.2. MIE as a dynamical fixed point

Proposition 1. *Let $\boldsymbol{x} = (\theta, b, \pi) \in \mathcal{X}$ denote the joint multilevel state that is within a nonempty, compact, and convex subset of a finite-dimensional Euclidean space $\prod_{i=1}^N (\Theta_i \times \mathcal{B}_i \times \Pi_i)$, and let Φ denote the coupled mean update operator. If $\boldsymbol{x}^* = \Phi(\boldsymbol{x}^*)$, and the components of Φ correspond respectively to environmental stationarity, neural learning, cognitive updating, and behavioral best response, then $\boldsymbol{x}^*$ satisfies the neural, cognitive, and behavioral equilibrium conditions and is therefore an MIE. Furthermore, if the mean coupled update operator $\Phi : \mathcal{X} \to \mathcal{X}$ is continuous, then at least one MIE exists.*

Sketch of Proof. Because $\mathcal{X}$ is nonempty, compact, and convex and Φ is a continuous self-map, *Brouwer's Fixed-Point Theorem* guarantees that any continuous function mapping a compact, convex set to itself has at least one fixed point (Hatcher, 2002). This implies that there exists $\boldsymbol{x}^* \in \mathcal{X}$ such that $\boldsymbol{x}^* = \Phi(\boldsymbol{x}^*)$. From this definition, $\boldsymbol{x}^*$ is an MIE. □

Remark. Brouwer's theorem uses continuous single-valued functions, but best-response strategies in behavioral games are often set-valued. The existence of MIE as a generalized fixed point can be similarly proved based on *Kakutani's Fixed-Point Theorem* for general equilibrium (Hatcher, 2002).

Simple Counterexample: Failure under a discontinuous update. Let $\mathcal{X} = [0, 1]$ and define

$$\Phi(x) = \begin{cases} 1, & x < 0.5, \\ 0, & x \geq 0.5. \end{cases}$$

Although $\mathcal{X}$ is compact and convex, Φ has no fixed point. Indeed, $x < 0.5$ implies that $\Phi(x) = 1 \neq x$, whereas $x \geq 0.5$ implies $\Phi(x) = 0 \neq x$. Hence an MIE does not exist when the update map is discontinuous and does not admit an upper-hemicontinuous convex-valued extension.

3.4.3. Multilevel equilibrium principle

In a Markov game where each agent updates (i) neural parameters by stochastic approximation or MARL, (ii) beliefs by consistent filtering or probability prediction, and (iii) policies by a smooth mapping from (θ, b), any limit point $(\theta^\star, b^\star, \pi^\star)$ of the coupled dynamics that is stationary under the induced kernel is an MIE; conversely, any stable MIE corresponds to an attracting invariant set of the multilevel learning dynamics. This framework unifies human-human, AI-AI, and human-AI interactions and presents different conditions:

- **Within-level equilibrium:** fixed point of F only (cognitive), G only (neural), or best-response condition (behavior); this can apply to both human-human (e.g., F_i approximates Bayesian ToM, G_i embodies synaptic learning constraints) and AI-AI interactions (e.g., F_i is an explicit opponent model, G_i implements an error backpropagation).
- **Cross-level equilibrium:** MIE fixed point of the coupled operator.
- **Cross-agent equilibrium:** MIE applies to any mixture of humans and AI agents via heterogeneous operators.
- **Failure modes:** when no MIE exists, or when MIE exists but is unstable.

MIE generalizes the classical Nash equilibrium to intelligent systems with internal computation. Rather than being defined solely at the level of observable strategies, equilibrium arises when neural learning dynamics, cognitive belief models, and behavioral policies mutually stabilize across interacting agents.

3.5. Stability and and equilibrium selection

The definition of MIE characterizes equilibrium as a fixed point of coupled multilevel dynamics. In many interactive systems, however, multiple equilibria may exist. Understanding which equilibria are reached requires analyzing the *stability* of these fixed points under the learning and inference dynamics.

3.5.1. Multilevel dynamical system

Let $\boldsymbol{x}_t = (\theta_t, b_t, \pi_t)$ denote the global multilevel state of the system; the evolution of the system can be written as $\boldsymbol{x}_{t+1} = \Phi(\boldsymbol{x}_t)$, where Φ denotes the joint update operator induced by neural learning dynamics, belief updates, and policy adaptation. An MIE $\boldsymbol{x}^*$ is said to be *locally asymptotically stable* if there exists a neighborhood $\mathcal{U}$ of $\boldsymbol{x}^*$ such that for any initial condition $\boldsymbol{x}_0 \in \mathcal{U}$, the dynamical system converges to equilibrium: $\lim_{t\to\infty} \boldsymbol{x}_t = \boldsymbol{x}^*$. In stochastic learning systems, convergence may be defined in probability or almost surely.

3.5.2. Local stability under heterogeneous timescales

In the multilevel framework, stability depends jointly on neural adaptation rates, belief inference dynamics, and behavioral policy adaptation. Instabilities may arise when these processes operate on incompatible time scales or when feedback loops between agents amplify prediction errors. In contrast, a stable interaction emerges when multilevel learning dynamics converges to a common attractor in the joint state space, or simultaneously converge to attractors in their own state spaces.

Mathematically, the operators G, F, H couple processes operating on incommensurable timescales; it is therefore necessary to study local stability beyond traditional methods that are based on a mean Jacobian operator defined at the same timescale. Let $\boldsymbol{x} = (\theta, b, \pi)$ denote the global multilevel state. For simplicity, we consider discrete-time learning dynamics with the following Jacobian:

$$\mathbf{J} = D\bar{\Phi}(\boldsymbol{x}^*).$$

where D defines a differential operator on nonlinear mapping. Next, we rewrite the Jacobian in a form of block structure

$$\mathbf{J} = \begin{bmatrix} \partial_\theta \Phi_\theta & \partial_b \Phi_\theta & \partial_\pi \Phi_\theta \\ \partial_\theta \Phi_b & \partial_b \Phi_b & \partial_\pi \Phi_b \\ \partial_\theta \Phi_\pi & \partial_b \Phi_\pi & \partial_\pi \Phi_\pi \end{bmatrix}\Bigg|_{\boldsymbol{x}=\boldsymbol{x}^*}.$$

The diagonal blocks encode within-level sensitivity, whereas the off-diagonal blocks encode cross-level and cross-agent coupling. The equilibrium is locally asymptotically stable if $\rho(\mathbf{J}) < 1$; namely, the spectral radius (the maximum absolute value or modulus of its eigenvalues of $\mathbf{J}$) is smaller than unity.

To account for heterogeneous adaptation rates, consider a linear approximation

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t + \mathbf{A} f(\boldsymbol{x}_t),$$

where $\mathbf{A}$ is a block-diagonal matrix of agent- and level-specific adaptation rates. Then

$$\mathbf{J} = \mathbf{I} + \mathbf{A}\mathbf{K}, \qquad \mathbf{K} = Df(\boldsymbol{x}^*),$$

and stability requires $\rho(\mathbf{I} + \mathbf{A}\mathbf{K}) < 1$. Thus, adaptation rates affect stability jointly with the local coupling matrix $\mathbf{K}$; they cannot in general be analyzed independently. When adaptation rates are strongly separated, multiple timescales coexist; the spectral radius remains a valid binary criterion for the complete discrete-time map, but it can be dominated by eigenvalues close to one that correspond to extremely slow modes. To mitigate the dominance of slow modes, we can use other measures such as modal relaxation times and stability margins.

For sequential or asynchronous updates, the Jacobian must be constructed over a complete update cycle. If neural, cognitive, and behavioral operators are applied sequentially, then the joint Jacobian should be computed as

$$\mathbf{J}_{\text{cycle}} = \mathbf{J}_F \mathbf{J}_G \mathbf{J}_H,$$

and local stability requires $\rho(\mathbf{J}_{\text{cycle}}) < 1$.

In sum, the appropriate stability analysis depends on the temporal implementation of the coupled operators: a one-step Jacobian for synchronous discrete updates, a product or monodromy matrix for periodic multirate updates, and a reduced fast-slow analysis under strong timescale separation.

3.5.3. Equilibrium selection

When multiple equilibria exist, the equilibrium reached by the system depends on initial conditions and the geometry of the learning dynamics. Each equilibrium corresponds to an *attractor basin*:

$$\mathcal{B}(\boldsymbol{x}^*) = \{\boldsymbol{x}_0 : \lim_{t\to\infty} \boldsymbol{x}_t = \boldsymbol{x}^*\}.$$

Different equilibria may correspond to different modes of interaction, including cooperative, competitive, or unstable regimes.

Equilibrium selection can be based on human-defined criteria (such as empirical thresholds) about the loss or matching quantities. In human-machine shared driving applications, an LLM-based game equilibrium selection approach was proposed (Yang et al., 2026), followed by a human-in-the-loop experiment for strategy validation; the LLM based its decisions on scenario understanding and prior knowledge, flexibly adapting to different contexts and accurately grasping human-machine interactions.

4. Applications of multilevel game theory

4.1. Human-autonomous vehicle interaction as a multilevel game

Autonomous driving offers an excellent illustration of the multilevel equilibrium theory because the environment involves continuous strategic interaction between human drivers and artificial agents controlling autonomous vehicles (AVs) (Shu et al., 2023). Human drivers continuously predict the actions of others, while autonomous systems must infer human intent and adapt their control policies accordingly. Therefore, a vehicle–driver interaction forms a two-agent stochastic dynamic game, in which equilibrium corresponds to stable and predictable driving behavior emerging from coupled human cognition and AI decision-making.

Traditionally, this human-AV interaction can be framed as non-cooperative games (Chen et al., 2024): Both the human and the AV act in their own self-interest. The AV's algorithms predict human behavior and calculate the optimal trajectory, penalizing for collisions while rewarding efficiency. However, bidirectional interaction between human and AV would allow the AV to safely probe the human, enabling the AV to alter its own trajectory to influence the human's next move, forcing a cooperative equilibrium (Fisac et al., 2019).

4.1.1. Traffic dynamics

Consider a human driver H interacting with an autonomous vehicle A. Let $s_t \in \mathcal{S}$ denote the traffic state at time t, which may include positions, velocities, road geometry, and other relevant environmental variables. Each agent selects an action $a_t^H \in \mathcal{A}_H, a_t^A \in \mathcal{A}_A$, corresponding to control variables such as steering, acceleration, or braking. The joint action is $\boldsymbol{a}_t = (a_t^H, a_t^A)$. The traffic state evolves according to $s_{t+1} \sim P(\cdot \mid s_t, a_t^H, a_t^A)$.

4.1.2. Multilevel representation of the agents

Each agent is represented using the multilevel state formulation $\boldsymbol{x}_t^i = (\theta_t^i, b_t^i, \pi_t^i)$ for $i \in \{H, A\}$.

Neural level. For the human driver, θ_t^H represents neural processes related to perception, decision making, and motor control. For the autonomous vehicle, θ_t^A represents parameters of the perception and control algorithms (e.g., artificial neural network weights used in trajectory prediction).

Cognitive level. Both agents maintain internal models of each other's behavior based on priors or past driving experiences. The human driver forms beliefs about the autonomous vehicle $b_t^H \approx P(a^A \mid s)$, while the autonomous vehicle maintains a model of human intent $b_t^A \approx P(a^H \mid s)$.

Behavioral level. Each agent selects stochastic or deterministic actions according to the policy $a_t^H \sim \pi_t^H(s_t, b_t^H), \quad a_t^A \sim \pi_t^A(s_t, b_t^A)$. These policies determine observable driving behavior.

4.1.3. Multilevel interactive equilibrium in traffic

A human-AV interaction reaches an MIE when the tuple $(\theta_H^*, \theta_A^*, b_H^*, b_A^*, \pi_H^*, \pi_A^*)$ satisfies the equilibrium conditions: (i) the neural earning dynamics stabilizes for both agents: $\nabla_{\theta_H} J_H = 0, \quad \nabla_{\theta_A} J_A = 0$; (ii) both agents develop consistent beliefs or expectations about each other's behavior: $b_H^* \approx P(a^A \mid s), \quad b_A^* \approx P(a^H \mid s)$; (iii) two driving strategies (π_H^*, π_A^*) become mutually stable.

> Example. Consider a highway merging scenario where a human driver attempts to merge into a lane occupied by an autonomous vehicle. Initially, both agents are uncertain about each other's behavior. Over repeated interactions, the human driver learns that the autonomous vehicle tends to yield, while the autonomous system learns that the human merges once sufficient space becomes available. The interaction may converge to a stable behavioral pattern: the autonomous vehicle slows slightly to allow the merge while the human driver merges once the gap appears. This configuration represents a multilevel equilibrium supported by consistent neural expectations, cognitive belief models, and behavioral policies.

The same analogy also applies to homogeneous intelligent AV-AV interactions. Unlike human-AV interactions where information is asymmetric, every agent in AV-AV interactions possesses approximately the same computational architecture. Each vehicle maintains internal representations of the environment, predicts the intentions of neighboring vehicles, updates its planning and control policies, and communicates either explicitly (e.g., V2V wireless communication) or implicitly through driving behavior. Consequently, all agents evolve under comparable computational principles, although they may employ different algorithms. Conventional Nash equilibrium concepts typically characterize only the final driving behavior, whereas MIE additionally seeks convergence of the internal predictive models and belief states that generate these behaviors. An MIE is achieved when the perception models, intention estimates, planning strategies, and driving policies of all vehicles become mutually consistent, allowing coordinated behaviors such as merging, platooning, or intersection negotiation to emerge.

4.2. Brain-machine interaction as a two-agent game

BMIs can be naturally modeled as an interactive system involving two co-adaptive agents: a biological brain and an artificial decoding system. Within the multilevel game framework, the brain and machine jointly form a coupled dynamical system whose behavior emerges from the interaction between neural activity, decoder control, and task dynamics.

4.2.1. Setting

The BMI system consisting of a biological brain H and an artificial agent M (e.g., machine decoder). The internal state of the brain is represented by $\boldsymbol{x}_t^H = (\theta_t^H, b_t^H, \pi_t^H)$, where

- θ_t^H denotes neural circuit parameters or plasticity states,
- b_t^H represents internal beliefs about the decoder mapping,
- π_t^H represents the neural policy that generates neural activity patterns n_t: $n_t \sim \pi_t^H(s_t, b_t^H)$.

The machine decoder (such as a Kalman filer or a neural network) is represented by $\boldsymbol{x}_t^M = (\theta_t^M, b_t^M, \pi_t^M)$, where

- θ_t^M denotes decoder parameters,
- b_t^M represents the machine's model of user intent,
- π_t^M denotes the control policy that maps neural activity to device commands u_t: $u_t \sim \pi_t^M(n_t, b_t^M)$.

Under this formulation, brain-machine interaction can be viewed as a two-agent multilevel game in which neural learning, belief inference, and behavioral control policies co-evolve. Successful BMI control corresponds to a multilevel equilibrium between a biological brain and a co-adaptive artificial agent. Stable BMI control emerges when the coupled dynamics between the brain and the decoder converge toward an MIE; namely, three processes converge simultaneously: (i) neural plasticity in the brain; (ii) learning in the machine decoder; and (iii) mutual prediction between the brain and machine. On the other hand, BMI instability may imply the failure of equilibrium, when either neural adaptation and decoder learning operate on incompatible timescales, or belief models fail to converge. This can lead to nonstationary co-adaptation.

This theory also makes several experimental predictions. First, shared internal models may improve equilibrium. For example, if the machine explicitly models the intent of the user $b_M \approx b_H$, the equilibrium may be reached faster. This motivates intent-aware decoders and co-adaptive BMI algorithms (Madduri et al., 2021, 2024).

Another prediction is that the system is stable if the machine's adaptation rate is much smaller than the human's adaptation rate (i.e., $\alpha_M \ll \alpha_H$), which matches empirical BMI findings where slow decoder updates stabilize learning (Oweiss & Badreldin, 2015). To formalize this as an experimental prediction, Let α_H denote the effective neural adaptation rate (e.g., trial-to-trial change in neural tuning or performance attributable to the user), and let α_M denote the decoder adaptation rate (e.g., step size or update magnitude in the decoder algorithm); let ℓ denote normalized decoder accuracy. One working hypothesis is that successful co-adaptation is governed not by a fixed value of α_M/α_H, but by a stability boundary that depends on decoder performance. Mathematically, it is written as

$$\frac{\alpha_M}{\alpha_H} \leq \kappa(\ell)$$

where $\kappa(\ell)$ denotes a decreasing function of decoder accuracy. Intuitively, when decoder accuracy is low, faster machine adaptation can be tolerated; as accuracy improves, overly rapid decoder updates may

destabilize the learned neural mappings. Under a simple first-order linearized approximation of the co-adaptation model, we rewrite the above equation as

$$\frac{\alpha_M}{\alpha_H} \leq c(1-\ell)$$

where c is a task-dependent constant that reflects decoder noise, neural variability, and task complexity. This prediction has an intuitive interpretation. During early learning, when decoder accuracy is relatively low, faster decoder adaptation can be tolerated because the decoder still identifies the appropriate neural manifold. As accuracy ℓ improves, however, overly rapid decoder adaptation continually changes the decoding map experienced by the user, preventing stable neural consolidation. Consequently, the admissible ratio α_M/α_H decreases as decoder accuracy approaches its asymptotic value. This also predicts a trade-off between stability and convergence: $T_{\text{conv}} \approx \mathcal{O}(1/\min\{\alpha_H, \alpha_M\})$ subject to stability constraint, where T_{conv} denote the time required for the joint system to reach a stable performance plateau.

This formulation is directly falsifiable. Existing co-adaptive BMI studies routinely report decoder accuracy trajectories, decoder update parameters, and behavioral learning curves. From these data researchers can estimate $\alpha_H, \alpha_M, \ell(t)$, and T_{conv}, and test whether stable systems satisfy the predicted inequality. Observation of stable convergence for adaptation ratios substantially exceeding the predicted stability boundary, or failure of the predicted relationship between $\ell(t)$ and allowable decoder adaptation rate, would falsify the proposed timescale hypothesis.

4.2.2. Mixed timescale adaptation

In BMI systems, decoder adaptation should respect the neural learning timescales. Here we briefly outline a mathematical framework to formulate mixed timescale adaptation using multirate stochastic approximation.

Without loss of generality, we assume that neural or decoder dynamics evolve on the fastest timescale, cognitive states evolve on an intermediate timescale, and overt behavioral policies evolve on the slowest timescale. For a BMI, for instance, a decoder may update on every trial, human beliefs over several trials, and neural representations across sessions. Conversely, rapid human sensorimotor adaptation may make the decoder tune for the slower subsystem. In this case, each slower process treats the faster processes as approximately equilibrated, while each faster process treats the slower variables as quasi-static: $\lambda_\theta(b^*, \pi^*)$, which together form a candidate MIE for the brain, denoted as $H\left(\lambda_\theta(\lambda_b(\pi), \pi), \lambda_b(\pi), \pi\right)$. This implies that the selected decoder's learning rate should preserve stability of the fast boundary-layer dynamics and of the reduced slow dynamics. If the decoder updates too aggressively, the assumed attracting manifold may cease to exist, lose stability, or move faster than the biological subsystem can track. If it updates too slowly, the joint system may remain stable but exhibit large tracking error or prolonged nonstationarity.

To make this mathematically explicit, let w denote decoder parameters and z denote biological neural states:

$$\dot{z} = f(z, w), \qquad \dot{w} = \eta g(z, w).$$

where $f(\cdot)$ and $g(\cdot)$ are two generic nonlinear systems. When $0 < \eta \ll 1$, the decoder regards the neural subsystem as approximately equilibrated at $z = h(w)$, and its reduced dynamics are

$$\dot{w} = \eta g(h(w), w).$$

This slow-decoder approximation is justified only if, for each relevant w, the neural boundary-layer system $\dot{z} = f(z, w)$ has a uniformly attracting equilibrium or invariant set. Alternatively, if the decoder is the fast process, the ordering can be reversed, but the decoder's conditional equilibrium should remain stable for quasi-static neural states.

Although timescale separation does not guarantee the existence of local stability of MIE, it becomes analytically useful when combined with stability of the fast boundary-layer systems, stability of the reduced slower systems, boundedness, regularity of the equilibrium manifolds, and suitable stochastic-approximation conditions. In simple evolutionary games, oscillatory dynamics are suppressed by heterogeneous adaptation rates of players (Masuda, 2008). For a systematic review on evolutionary multi-agent coordination under heterogeneous adaptation rates, see Wang et al. (2025).

4.3. Human-LLM interaction

Interactions between humans and LLMs such as ChatGPT can be interpreted as a two-agent interactive system. In this setting, a human user and a language model exchange messages through a conversational interface. The dialog evolves as a sequential decision process in which both agents adapt their responses based on the evolving conversational context. To date, prior human-LLM interaction research has mainly focused on the classical Nash-equilibrium (Akata et al., 2025; Barak & Costa-Gomes, 2025; Zhu, 2025b).

4.3.1. Conversation dynamics

Let $s_t \in S$ denote the conversational state at time t, which can represent the dialog history, including previous prompts, responses, and contextual information. The human agent H produces a prompt $p_t \in \mathcal{P}$, and the language model agent M produces a response $r_t \in \mathcal{R}$. The conversational state evolves according to

$$s_{t+1} = f(s_t, p_t, r_t),$$

where f denotes the dialog update operator that appends new messages to the conversation context.

4.3.2. Multilevel agent representation

Within the multilevel framework, both agents are represented by neural, cognitive, and behavioral states.

Human agent. The human internal state is represented as $x_t^H = (\theta_t^H, b_t^H, \pi_t^H)$, where

- θ_t^H represents the neural processes that underlie reasoning, language production, and learning,
- b_t^H represents the user's internal belief about the LLM's capabilities and behavior,
- π_t^H represents the prompting strategy used by the user.

The human generates prompts according to the policy

$$p_t \sim \pi_t^H(s_t, b_t^H).$$

LLM agent. The language model agent is represented by $x_t^M = (\theta_t^M, b_t^M, \pi_t^M)$, where

- θ_t^M denotes the parameters of the language model,
- b_t^M represents the model's inferred representation of user intent,
- π_t^M denotes the response generation policy.

Responses are generated according to the policy

$$r_t \sim \pi_t^M(s_t, b_t^M).$$

4.3.3. Belief and learning updates

Both agents update their internal representations through bidirectional interaction. The human updates expectations about the LLM based on observed responses,

$$b_{t+1}^H = F_H(b_t^H, r_t),$$

while the LLM updates its internal representation of user intent through contextual inference,

$$b_{t+1}^M = F_M(b_t^M, p_t).$$

In addition, learning may occur at longer timescales through neural adaptation in the human brain or through parameter updates in the LLM.

Therefore, human-LLM dialog can be viewed as a multilevel interactive game in which conversational strategies emerge from the coupled dynamics of neural cognition, belief inference, and language generation policies. Efficient communication arises when the interaction converges toward an MIE in which both the human's prompting strategy and the model's response policy become mutually predictable and stable. Prompt engineering reduces uncertainty in belief estimation and accelerates convergence, whereas failure of cognitive equilibrium generates misunderstanding or one type of LLM hallucination due to dialog-induced belief misalignment. Specifically, hallucination is treated as an equilibrium failure only when it reflects instability or inconsistency in the coupled dynamics between internal state inference and conversational context. This is in contrast to another type of hallucination arising from intrinsic generative miscalibration that is outside the MIE framework. Preliminary theoretical analyses of behavioral equilibria in LLM-Nash games have been discussed in Zhu (2025b).

4.4. Psychiatric disorders as disrupted multilevel equilibria

Psychiatric disorders with dysfunctional social interactions can be studied in the perspective of multi-agent game theory (MAGT) (Chen, 2026). In the multilevel framework, a healthy individual interacting with the environment can be described by the state (θ, b, π), where θ represents neural parameters and circuit dynamics; b represents cognitive beliefs or internal models of the world and other agents; π represents behavioral policies that govern action. Healthy functioning corresponds to a stable multilevel equilibrium $(\theta^\star, b^\star, \pi^\star)$, in which neural learning, cognitive inference, and behavioral strategies remain mutually consistent with environmental feedback. Therefore, from the computational psychiatry viewpoint, a working hypothesis is that psychiatric disorders can arise when this equilibrium is distorted, unstable, or trapped in maladaptive attractor states.

4.4.1. Three types of pathological equilibria

According to the multilevel equilibrium framework, we define a "pathological equilibrium" as a stable, self-reinforcing attractor in the joint neural-cognitive-behavioral state space that satisfies three conditions: (i) The joint state remains stable under small exogenous perturbations and returns to the same attractor after transient deviations. (ii) Neural, cognitive, and behavioral variables mutually reinforce one another, forming a closed feedback loop. (iii) The equilibrium is associated with measurable deficits in adaptive functioning relative to normative baselines. Based on different levels of dysfunction, we envision that three types of pathological equilibria may occur.

- *Neural-level instability*: At the neural level, abnormal circuit dynamics or neurotransmitter signaling can destabilize learning processes. Examples of neural mechanisms include altered dopaminergic prediction errors, abnormal cortical excitation-inhibition (E/I) balance, and impaired synaptic plasticity. Formally, neural dynamics G may (or may not) converge to pathological attractors, which can further propagate upward to cognitive and behavioral dysfunction.
- *Cognitive-level miscalibration*: At the cognitive level, disorders may involve incorrect belief updates about the environment or other agents, reflecting social dysfunctions in trust or representing positive symptoms. If beliefs evolve as $b_{t+1} = F(b_t, o_t)$, pathology may arise when $b^\star \neq \mathbb{E}[F(b^\star, o)]$ or when beliefs converge to incorrect fixed points. Examples of such behavior include exaggerated threat expectations, incorrect inference of social intent, and persistent negative self-beliefs.
- *Behavioral-level maladaptive strategies*: At the behavioral level, individuals can adopt strategies that are locally stable but globally maladaptive. For example, $\pi^\star \neq \arg\max V(\pi)$ with respect to long-term well-being. These policies may remain stable because they are reinforced by distorted beliefs or abnormal neural signals.

Note that pathological equilibria distinguish maladaptive equilibria from transient dynamic regimes (which exhibit nonstationary trajectories without convergence) and from multistability (in which multiple attractors coexist and transitions occur due to noise or external shocks). In the case of comorbidity in psychiatric disorders, the framework also accommodates multistability by allowing multiple locally stable equilibria corresponding to different symptom clusters; empirical identification then requires detecting switching dynamics or basin structure rather than assuming a single fixed point.

4.4.2. Examples of psychiatric phenotypes

Under the proposed framework, many psychiatric disorders can be interpreted as pathological equilibria.

- **Depression**, which may correspond to a negative cognitive equilibrium. At the neural level, it manifests itself as reduced dopaminergic reward prediction signals alter learning dynamics. At the cognitive level, beliefs about future reward become pessimistic (i.e., $b^\star$(reward)$\ll$ true reward probability). At the behavioral level, individuals adopt withdrawal strategies; consequently, the system stabilizes in a low-reward equilibrium.
- **Anxiety disorder**, which may correspond to a threat-biased equilibrium. In this case, cognitive beliefs overestimate risk (i.e., $b^\star$(reward)$\gg$ true reward probability), or behavioral strategies become more avoidance-based. Although avoidance temporarily reduces perceived risk, it prevents corrective evidence and reinforces the equilibrium.
- **Schizophrenia or psychosis**, which may involve unstable belief dynamics. At the cognitive level, aberrant prediction errors lead to excessive belief updates, which can furtehr generatedelusional beliefs, hallucinations, or impaired ToM models.
- **Autism spectrum disorders**, which may involve altered equilibrium in social interactive games. Differences in social belief modeling may lead to mismatches between internal models of other agents, or observed social behavior. As a result, this can produce difficulty of stabilizing social interaction equilibria.

It is worth emphasizing that social interaction is a defining feature of dynamic human behavior, emerging from complex cognitive processes that enable individuals to infer others' intentions, adapt their actions, and coordinate in dynamic environments. Disruptions in social interaction are central to a range of psychiatric conditions, characterized by deficits across neural, cognitive and behavioral levels. The MIE is therefore a natural framework to characterize interactions among (healthy or mentally ill) persons and define the emergent properties from interactive behaviors. More discussions are referred to Chen (2026).

4.4.3. Implications for treatment in precision psychiatry

As a useful hypothesis-generating application, the new framework also suggests treatment protocol by perturbing the system across levels. Namely, interventions can be conceptualized as targeted perturbations to specific levels. For example, at the neural level, pharmacotherapy modulate neural dynamics by targeting neural parameters θ (e.g., fMRI connectivity patterns or EEG spectral features); at the cognitive level, psychotherapy or cognitive-behavioral interventions targets cognitive update b (e.g., belief expectation, uncertainty, or valuation); at the behavioral level, behavioral interventions modify policy π in controlled tasks or ecological monitoring. Accordingly, successful treatment may aim to push the system out of a pathological attractor toward a healthier equilibrium. However, identification of these neural, cognitive, and behavioral parameters is subject to statistical uncertainty. Nevertheless, the MIE framework provides a principled way to predict how such perturbations may shift the system between attractors. To date, individualized equilibrium parameters lack a concrete protocol for parameter identification, the modeling hypothesis thus still requires rigorous empirical grounding and validation in clinical practice.

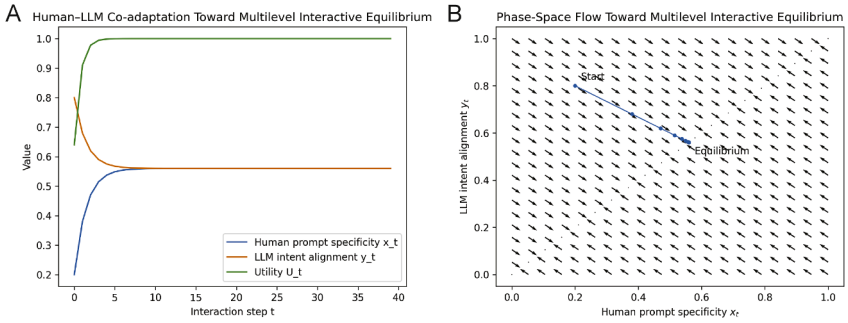


Fig. 2. A toy interaction model illustrates how prompt specificity x_t (human strategy) and intent alignment y_t (LLM inference) evolve through coupled adaptation dynamics. (A) Task utility U_t increases as the mismatch between human prompting and model interpretation decreases. The system converges toward a fixed point or MIE where $x_t \approx y_t$. (B) The phase-space flow in (x_t, y_t) space shows the coupled evolution of human prompt specificity x_t and LLM intent alignment y_t. The blue trajectory shows convergence from an initial mismatch toward a stable fixed point along the diagonal.

5. Case studies via computer simulations

5.1. Human-LLM co-adaptation

Consider a simple scenario: a human repeatedly asks an LLM to summarize scientific papers. Initially, the human uses underspecified prompts such as "Summarize this," while the LLM responds generically. After repeated interactions, the user learns to provide more structured prompts, such as "Summarize the main contribution, key methods, and limitations in three bullet points." In parallel, the LLM infers the user's preferred output format from context. Over time, prompt specificity and model alignment converge, task utility increases, and the interaction stabilizes into a multilevel equilibrium characterized by efficient and predictable communication. Fig. 2 presents a simple illustrated example derived from computer simulation (see Appendix A for details).

5.2. AI-AI interactions in trust games

MIE can be empirically tested by computer simulations. Here we consider a trust game experiment between two AI agents (Appendix B). In this gameplay, Agent 1 (Investor) and Agent 2 (Trustee) are modeled by two gated recurrent neural networks (RNNs), which aim to maximize their reward or minimize the loss (Fig. 3A, see Appendix B for details). To introduce non-stationarity, we assume that Investor has two policies A and B) and switch its strategy twice during the repeated game ($A \rightarrow B$ and then $B \rightarrow A$). As a result, the Trustee needs to adapt its policy based on the loss or reward in the feedback. Traditionally, the behavioral equilibrium is measured via the observed outcome (reward, loss), which can be studied within the MARL framework using model-free actor-critic RL (Fig. 3B). Here, we generalize the concept by studying neural and cognitive equilibria, at both within-level and cross-level (Fig. 3C; see Appendix B for details). In our experiments, we characterize the RNN's population activity via principal component analysis (PCA) and identify fixed-point attractors that are related to pre-specified policies (see animation demo in the github), and measure the minimum distance of population activity to the fixed points in the neural space. Alternatively, we calculate the dominant canonical correlation (CC) between population activities of two RNNs in the shared neural subspace. Additionally, the agent RNN output predicts the opponent's action of the next round, the prediction error between the predicted and actual action can be used as an empirical criterion for the cognitive equilibrium. When neural, cognitive, and behavioral equilibria are simultaneously satisfied, MIE arises (Fig. 3C).

As a possible application demonstration, we envision that acute stress can affect trust priors and regulate social decision-making. In computer simulations, we modify the learning-rate parameter of Trustee agent as a function of stress level while keeping the learning-rate of Investor agent as a constant (Appendix B). We further compare the convergence time-to-equilibrium under different relative stress level (range: 0–1; 0 being no stress). As noted, the stress level positively correlate with the delay to equilibria across levels, but the MIE achieves the highest Pearson's correlation and the lowest P-value (Fig. 3D). Collectively, these experiments provide a proof-of-concept to study MIE in social interactions and its relationship to mental health (e.g., stress, anxiety, depression). A systematical study of MAGT and MIE for precision psychiatry as well as detailed discussions will be reported elsewhere.

6. Estimating MIE: experimental strategies and computational methods

The MIE theory defines equilibrium as a fixed point of coupled neural, cognitive, and behavioral dynamics. However, the theory does not directly provide a practical means to estimate or identify the equilibrium. In practice, these quantities are rarely directly observable. Instead, equilibrium must be estimated from empirical interactional data through a combination of behavioral analysis, latent-state inference, and dynamical system modeling (e.g., based on artificial neural networks).

6.1. Neural dynamics and stationarity

At the neural level, equilibrium corresponds to stationary learning dynamics. Neural signals (e.g., population activity or prediction errors) can be analyzed using dynamical system models. Neural equilibrium can be determined via visualization or dimensionality-reduction methods. Existing latent dynamical system techniques, such as factor analysis, state-space modeling, or neural manifold learning, can be used to identify attractor structures in neural population activity.

One approach is to analyze the latent state trajectory using attractor analysis. A fixed-point, line attractor, limit cycle, or periodic oscillator may reveal stationary patterns of neural dynamics at different timescales. Another approach is neural decoding of belief and policies. In the BMI study, one can also train decoders from neural activity to predict the choice or the opponent's action belief, and test whether these decoded variables stabilize across repeated interactions.

We can also jointly examine neural activities of two agents. One possible approach is to linearly project them separately into a low-dimensional subspace that maximizes their correlation (e.g., canonical correlation analysis or CCA), or maximize their covariance matrix by partial least-squared correlation (PLSC) (Jiang et al., 2026; Zhang et al., 2025). Once the shared component in the subspace is identified, we can monitor the convergence of neural population trajectories of two agents projected separately in the dominant shared subspace.

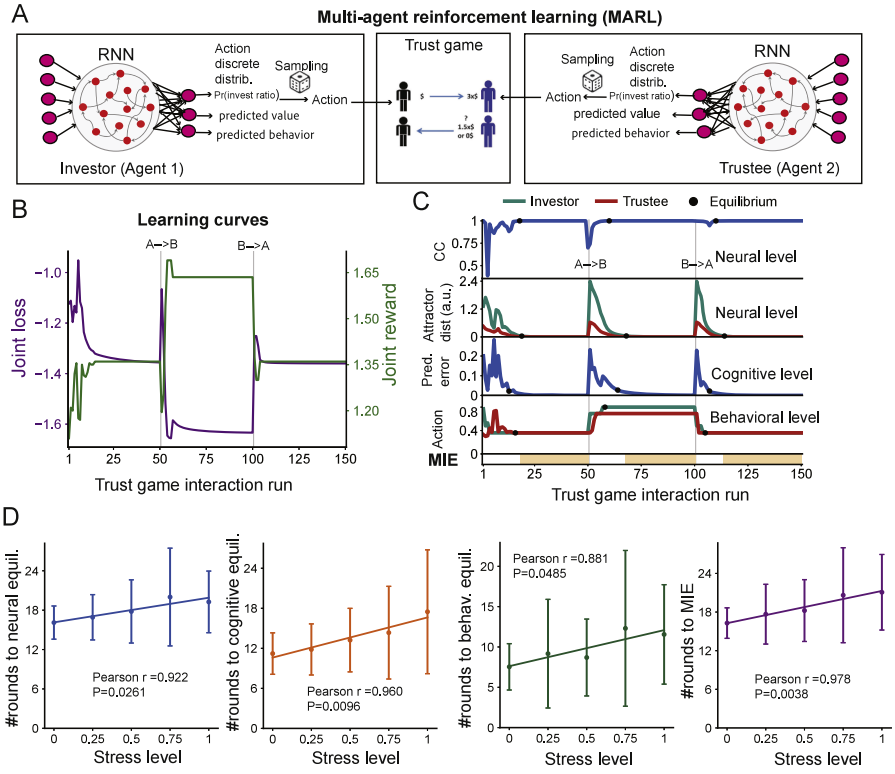


Fig. 3. (A) Schematic of multi-agent reinforcement learning (MARL) and RNNs in an AI-AI Trust game. (B) Representative learning curves (shown in joint reward or joint loss from RNNs) in 150-run trust game interactions with non-stationarity, where the Investor's policy switches twice ($A \to B$ and $B \to A$). (C) Monitoring neural (attractor-distance based or canonical correlation-based), cognitive, and behavioral equilibria based on the proposed metric and threshold criteria separately. At the bottom, MIE corresponds to the overlapping period when three equilibria simultaneously occur. (D) The relative stress level of the Trustee with the convergence time toward neural, cognitive, behavioral equilibria and MIE. Higher stress in the Trustee negatively affects the mutual trust, or the time that reaches to equilibria. Pearson's correlation r and P-value are reported based on Monte Carlo simulations (n = 26 per stress condition; error bar denotes SD).

6.2. Cognitive state estimation

Cognitive belief states are typically latent and must be inferred from the observed behavior. A common approach is to model beliefs as hidden states in a state-space model (SSM). The multilevel equilibrium is then estimated as a fixed point of the inferred dynamical system. Belief trajectories can then be estimated using Kalman filtering, approximate Bayesian filtering, or variational inference. Cognitive equilibrium is suggested when inferred beliefs stabilize,

$$\|\hat{b}^i_{t+1} - \hat{b}^i_t\| < \epsilon.$$

Experimental probes, such as eliciting explicit predictions about an opponent's actions, can further validate inferred belief states.

One can also design experimental probes to reveal each agent's internal model of the other. For humans, this opponent-modeling strategy includes predicting the opponent's next move, collecting trial-by-trial confidence estimates. For AI systems, this strategy includes estimating the inferred user's intent, analyzing attention to the user history, and fitting interpretable surrogate models. Next, human or AI agents then compare the inferred beliefs $\hat{b}_i$ with the observed opponent policy $\hat{\pi}_{-i}$ (using Kullback-Leibler divergence or Wasserstein distance).

Another strategy involves recursive belief modeling or iterative reasoning. If agent i represents belief about other agent's belief as $b_i(a_j)$, first-order beliefs can be "what I think you will do," and second-order beliefs can be "what I think you think I will do" and so on, yielding a chain of chain of thoughts (Zhu, 2026). In human-LLM interaction or psychiatric social inference experiment, one can fit nested belief models such as $b^{(i)}_1$ = *belief about other's action* and $b^{(i)}_2$ = *belief about other's belief*, then compare these two models of different depth using held-out prediction.

6.3. Behavioral estimation

At the behavioral level, one can run repeated trials of the same interaction trials (such as closed-loop BMI control, human-LLM iterative prompting). Given observed history of state-action frequencies, an empirical policy estimate may be obtained as

$$\hat{\pi}_i(a \mid s) = \frac{N_i(s,a)}{\sum_{a'} N_i(s,a')},$$

where $N_i(s,a)$ denotes the number of times agent i selects action a in state s. Behavioral equilibrium may be achieved when policy drift becomes small over time for all agents over a sustained interaction window:

$$\|\hat{\pi}_{i,t+1} - \hat{\pi}_{i,t}\| < \epsilon.$$

An additional diagnostic is the *best-response gap* (Pangallo et al., 2019): given the estimated opponent policy $\hat{\pi}_{-i}$, compute whether the observed policy is close to a best response:

$$\mathrm{BRGap}_i = \max_{\pi_i} V_i(\pi_i, \hat{\pi}_{-i}) - V_i(\hat{\pi}_i, \hat{\pi}_{-i}),$$

which measures the incentive for unilateral deviation. If BRGap is sufficiently small for all agents, behavior is near equilibrium.

For a given stochastic dynamical policy model, $\pi_{t+1} = f_\phi(\pi_t, s_t, o_t)$, one can also estimate its fixed points and examine its behavioral dynamics.

6.4. Joint estimation

A principled strategy is to jointly model neural, cognitive, and behavioral processes using hierarchical state-space models,

$$\theta_{t+1} = G(\theta_t, \delta_t) + \xi_t^{\theta},$$

$$b_{t+1} = F(b_t, o_t) + \xi_t^{b},$$

$$a_t \sim \pi(\cdot \mid s_t, b_t, \theta_t).$$

where ξ^{θ} and ξ^{b} denote the additive noise terms for θ and b system equations, respectively. Such models can be fitted using expectation-maximization (EM), variational Bayes, or particle Markov chain Monte Carlo (MCMC) methods. Alternatively, if two agents are represented by two artificial neural networks; the model parameters can be estimated based on gradient descent or policy gradient, followed by network-level attractor analyses (neural level), belief prediction error analysis (cognitive level), and policy comparison (behavioral level).

An empirical metric for the "distance to equilibrium" is

$$\mathcal{E}_t = \lambda_\theta \|\theta_{t+1} - \theta_t\| + \lambda_b \|b_{t+1} - b_t\| + \lambda_\pi \|\pi_{t+1} - \pi_t\| + \lambda_{\text{BRGap}} \sum_i \text{BRGap}_i,$$

where $\lambda_\theta, \lambda_b, \lambda_\pi$ and λ_{BRGap} denote the weighting parameters measuring the trade-off. When $\mathcal{E}_t$ becomes small and stays small, the system is near multilevel equilibrium. Alternatively, a normalized version can be adopted based on a dimensionless metric

$$\mathcal{E}_t = w_\theta \frac{\|\theta_{t+1} - \theta_t\|}{\|\theta_t\|} + w_b \frac{\|b_{t+1} - b_t\|}{\|b_t\|} + w_\pi \frac{\|\pi_{t+1} - \pi_t\|}{\|\pi_t\|} + w_{\text{normBRGap}} \sum_i \text{normBRGap}_i,$$

where normBRGap denotes the normalized best-response gap, w_θ, w_b, w_π and $w_{\text{normBRGap}}$ denote the respective weights that can be either set uniformly or learned via cross-validation to optimize predictive performance. Furthermore, sensitivity analysis can be conducted over the weighting scheme and test the robustness of a broad range of weight configurations and their impact on the MIE convergence.

6.5. Experimental validation through perturbation

Equilibrium stability can be experimentally tested through controlled perturbations. After an equilibrium is reached, perturbations may be introduced by modifying environmental conditions, altering agent policies, or injecting neural or cognitive perturbations. A stable MIE is expected to exhibit recovery toward the same fixed point, whereas instability may lead to transitions toward alternative attractors.

Since a true equilibrium should be resistant to small perturbations. To test local stability experimentally, after apparent convergence, one may perturb one layer, such as change reward contingencies, or alter AI response style, or modify decoder mapping, or inject misleading social cues, or perturb neural stimulation, then test whether the system returns to the same equilibrium, or moves to a new one, or becomes unstable. Furthermore, manipulation of timescale such as the adaptation rates (e.g., fast vs slow; fixed vs. adaptive) of one agent or one level may allow us to test the theory's prediction that equilibrium depends on compatible cross-level learning timescales. Additionally, at the cognitive level, one can consider masking or distorting information about the opponent in social interactions (or feedback information in BMI systems) to test how belief estimation affects equilibrium.

In RNN-based agent modeling during cooperative or competitive games, we can also estimate the shared neural subspace of two RNNs' population activities and further perturb the subspace to probe the behavioral impact (Jiang et al., 2026; Zhang et al., 2025).

7. Discussion

7.1. Relation to other work

7.1.1. Evolutionary game theory

There is a strong conceptual connection between MIE and evolutionary game theory (EGT) and evolutionary game dynamics (EGD) (Hofbauer & Sigmund, 1998; Sandholm, 2010), but they answer different questions and operate at different levels of abstraction. EGT specifies how behavioral strategies evolve under repeated interactions, whereas MIE characterizes the equilibrium of the coupled neural, cognitive, and behavioral dynamics that generate those strategies. First, EGT defines the dynamics trajectories (e.g., replicator dynamics, best-response dynamics, logit dynamics, RL, etc.) and describes how strategies evolve over time, yet MIE focuses on the equilibrium, or asks "When will the entire multilevel system stop changing"? Traditional evolutionary games evolve only policy π, whereas everything else is hidden or implicit. If π is chosen to be replicator dynamics, then EGD produces an evolutionary stable strategy (ESS) or evolutionary equilibrium as a special case (Maynard Smith, 1982): a degenerate MIE with frozen internal states. Second, MIE is agnostic about the behavioral learning rule, and allows flexible or adaptive payoff functions that are subject to the influences of beliefs, neural plasticity, attention or context, whereas most EGT assumes a fixed payoff function. Third, evolutionary dynamics studies strategy adaptation, whereas MIE studies co-adaptation of the brain, cognition, behavior and environment and allows (θ, b, π) evolve simultaneously. Therefore, The proposed MIE framework should be viewed as complementary to EGD.

7.1.2. MARL

The MIE framework is also closely related to multi-agent reinforcement learning (MARL), but differs fundamentally in the object of learning and the notion of equilibrium. In MARL, agents iteratively optimize their policies or value functions through RL updates, such as Q-learning or policy-gradient methods, until reaching a stationary point of the learning dynamics (Sutton & Barto, 2018). Mathematically, MARL focuses on the evolution of policy parameters (or value functions), whereas MIE explicitly models the coupled evolution of neural representations, cognitive states, and behavioral policies through the multilevel operators G, F, and H. Consequently, MIE defines equilibrium as a joint fixed point (θ^*, b^*, π^*) that is simultaneously stable across neural, cognitive, and behavioral levels, rather than only requiring convergence of behavioral policies. From a mathematical perspective, MARL can be viewed as a special case of the MIE framework. If cognitive states are omitted or absorbed into the environment state, neural variables correspond solely to policy or value-function parameters, and behavioral policies are fully determined by these parameters, then the MIE dynamics reduce to the standard stochastic approximation or policy gradient equations used in MARL. Under these assumptions, the MIE fixed-point conditions coincide with the stationary conditions of MARL. Therefore, MIE generalizes MARL by incorporating latent internal states, memory, and heterogeneous timescales into the learning dynamics, providing a unified framework for analyzing co-adaptation among biological agents, artificial agents, and human–AI systems where learning occurs simultaneously across multiple interacting levels.

MIE can be treated as a generalized model-free MARL. For instance, if we use RNNs to model the "actor" and "critic" in model-free RL, where the actor models behavioral policy and critic models cognitive belief, and the neural states are characterized by the internal RNN states. Therefore, MIE resembles a three-level actor-critic architecture.

7.1.3. Other established equilibrium concepts

MIE should be distinguished from, but can incorporate, established generalizations of Nash equilibrium (Nash, 1950). In addition to evolutionary equilibrium and MARL, Bayesian equilibria model uncertainty

Table 1
Comparison of MIE with representative equilibrium and learning frameworks.

Framework	Primary equilibrium object	Bounded rationality or uncertainty	Learning dynamics	Explicit internal (neural) state	Cross-level consistency
Nash equilibrium	Strategy profile	No	No	No	No
Bayesian Nash equilibrium	Strategies and beliefs over types	Yes	No	Beliefs only	Partial
Quantal response equilibrium	Probabilistic strategy profile	Yes	No	No	No
Repeated and stochastic games	History- or state-dependent policies	Possible	Possible	No	No
Self-confirming equilibrium	Strategies with on-path beliefs	Yes	Possible	Beliefs only	Partial (behavioral-belief consistency)
Evolutionary equilibrium	Population strategy distribution	Implicit	Yes	No	No
MARL	Policies, value functions, policy parameters	Yes	Yes	Algorithmic parameters	No
MIE	Neural, cognitive, and behavioral states	Yes	Yes	Yes	Yes

about types or states (Harsanyi, 1967–1968); quantal response equilibria represent bounded or noisy rationality (Goeree et al., 2016; McKelvey & Palfrey, 1995); repeated- and stochastic-game equilibria capture temporal and history-dependent interaction (Camerer, 2003); self-confirming equilibria permit beliefs that are accurate only along the realized path (Fudenberg & Levine, 1993). Recent reviews organize these approaches across static, dynamic, repeated, evolutionary, and learning-based games and emphasize the roles of uncertainty, behavioral heterogeneity, and adaptive coordination (Cheng et al., 2026).

Established equilibrium notions can be embedded through particular choices of the behavioral and cognitive operators. For example, a best-response operator yields a Nash-type behavioral marginal, a quantal-response operator yields a bounded-rational marginal, a replicator operator yields evolutionary dynamics, and RL updates yield a multi-agent learning system. However, stationarity of the behavioral marginal alone does not imply a full MIE: internal representations or beliefs may continue to adapt even after observable behavior has stabilized. See Table 1 for a short summary of comparison between different equilibrium concepts.

Therefore, MIE is an orthogonal, multilevel extension rather than a replacement for existing solution concepts. It asks whether the internal computational processes that generate behavior have become mutually consistent with the resulting policies and interaction environment. This distinction allows systems with identical behavioral equilibria but different internal states to exhibit different perturbation responses, path dependence, or adaptation timescales. It also permits the empirical separation of behavioral equilibrium, marginal MIE, and full cross-level equilibrium.

7.2. Falsifiability

At a conceptual level, MIE makes a stronger claim than behavioral co-adaptation: it posits that stable behavioral equilibria are jointly determined by, and contingent upon, coupled dynamics in cognitive and neural states. This implies that behavioral equilibria should exhibit systematic dependence on internal representations and belief states, even when external incentives remain unchanged. There are three classes of empirical signatures that distinguish MIE from purely behavioral adaptation.

- *Cross-level hysteresis and path dependence:* Under MIE, transient perturbations to cognitive or neural states should induce persistent shifts in behavioral equilibria, even after external conditions are restored. For example, if beliefs are experimentally manipulated and subsequently corrected, MIE predicts that behavior may not revert to its original equilibrium if the underlying neural or representational states have adapted. In contrast, standard behavioral models without internal state dynamics predict reversion once payoffs are restored.
- *Dissociation between objective and perceived payoffs.* MIE predicts that behavior tracks endogenously constructed payoffs, not objective payoffs alone. Therefore, experimentally altering cognitive states (e.g., attention manipulation) should systematically shift behavior even when objective payoff structures are held fixed. A purely behavioral co-adaptation model without cognitive mediation cannot account for such systematic deviations unless augmented with ad hoc payoff transformations.
- *Lagged and asymmetric adaptation across levels.* Due to timescale separation, MIE predicts that neural, cognitive, and behavioral adjustments occur at different rates. This implies observable lag structures—for instance, rapid behavioral adaptation followed by slower consolidation in neural signals, or vice versa. Moreover, perturbations at one level should propagate asymmetrically across levels, producing transient mismatches (e.g., behavioral adapting faster than neural representations). Such multirate dynamics are not predicted by single-level adaptive models.

However, MIE is falsifiable because it makes cross-level predictions: it asserts that internal states are not epiphenomenal but causally and dynamically coupled to behavior. If such coupling cannot be empirically demonstrated, the framework should be rejected in favor of simpler models. Empirical evidence of negative results may include: (i) Behavioral dynamics are fully explained by observable payoff structures and past actions, with no additional predictive power from measured cognitive or neural states; (ii) Perturbations to cognitive or neural variables have no persistent effect on behavior once external incentives are restored. To operationalize falsifiability, we can design specific experimental paradigms with controlled perturbation: (i) neural perturbations through noninvasive neurostimulation or pharmacological interventions; (ii) cognitive perturbations via misinformation or feedback distortion; (iii) behavioral perturbations through changing payoff structure or adding action constraints.

7.3. Identifiability of the joint estimator

The inference of joint MIE parameters (θ, b, π) is ill-posed and remains a technical challenge from a partially observable Markov process (POMDP). It is important to assess the identifiability issue by conducting parameter recovery experiments with properly controlled computer simulations. In the controlled experiment, the recovery analysis can evaluate how the parameter bias change in temporal trajectories, or whether uncertainty or variance shrinks with increasing data samples, or how reduced models (e.g., behavioral-only or cognitive-behavioral models)

are compared to a three-level model in performance evaluation using predictive likelihood.

In real experimental settings, identifiability is inherently conditional, depending on the richness of data, the informativeness of the observation model, and the separation of latent dynamics. When operating in the regime of limited observability, only partial or "marginal" inference becomes feasible.

7.4. Toward a multilevel science of interactive intelligence

The multilevel equilibrium framework may provide a unifying perspective across a wide range of NeuroAI systems. It suggests that the study of strategic interaction should incorporate not only behavioral outcomes but also the internal computational mechanisms that generate them. It also provides a common framework for comparing biological and artificial intelligence, since both can be analyzed in terms of interacting levels of representation and learning dynamics. In BMI, equilibrium may correspond to a stable alignment between neural encoding and decoder policies. In conversational AI, human-LLM dialog may converge toward efficient communication strategies as users and models learn each other's behaviors. In autonomous driving, interactions between human drivers and automated systems may stabilize into socially compatible traffic conventions. Finally, it highlights the possibility that instability or dysfunction in social behavior—such as those observed in some psychiatric conditions—may arise from misalignment across levels of the system, where neural learning dynamics fail to support stable cognitive models or behavioral strategies.

An outstanding issue concerns the existence and uniqueness of multilevel equilibria. Classical game theory has established conditions under which Nash equilibria exist, but analogous results for multilevel dynamical systems remain largely unexplored. Specifically, the joint state space may admit multiple attractors corresponding to distinct behavioral regimes. Determining when such equilibria exist, how many may occur, and whether they are stable under realistic learning dynamics represents an important theoretical challenge. Another open problem involves timescale separation across levels. Neural learning, belief updating, and behavioral adaptation often operate on different temporal scales or in an asynchronous and cyclic manner. Understanding how equilibrium emerges when these processes interact asynchronously may require new analytical tools from stochastic dynamical systems and MARL. Furthermore, decision-making or actions in social interactions may be driven by mutually independent cognitive and non-cognitive components. How to incorporate non-cognitive components (such as cultural norm, group influence, impulsive drive) into the game-theoretical framework remains largely unexplored.

A common problem of MARL is the instability because of non-stationarity. Cutting-edge strategies with opponent-learning awareness (LOLA) can mitigate this problem (Foerster et al., 2018). Another promising approach is the embedded predictive intelligence approach (Meulemans, 2025), in which embedded Bayesian agents model self and others and distinguish "self-similar" agents from "decoupled" agents, where interactions with self-similar agents can lead to a coherent framework for MARL. Applying this framework to estimate cognitive or behavioral equilibrium may prove fruitful for rational games.

In practice, testing the multilevel equilibrium hypothesis requires measurements that simultaneously capture neural activity, cognitive inference, and behavior. With advances in large-scale brain recording and computational modeling, a key methodological challenge is the identification of latent cognitive variables, such as internal beliefs or opponent models, from behavioral data; this problem becomes more challenging when the number of interacting agents increases.

Several directions may be particularly promising for future work. First, development of computational tools for estimating MIE from real-world interaction data is critical. Advances in hierarchical state-space modeling and deep dynamical systems may enable joint estimation of neural, cognitive, and behavioral dynamics. Second, future research should explore collective multilevel equilibria in larger populations of interacting agents. Real-world systems such as social networks, traffic systems, human-robot and human-AI ecosystems, often involve many agents whose interactions generate emergent group-level behavior. Fourth, digital twins and MARL framework can provide insight into the theory and estimation of MIE given complete neural observability (Zhang et al., 2025); neural perturbation in silico can further causally test new hypotheses or suggest innovative human intervention strategies. Fifth, applying MAGT to psychiatry has gained growing interests (Chen, 2026; Cullen et al., 2018; Loula & Monteiro, 2020); combining game-behavior paradigms, cognitive tasks, and EEG/fMRI neuroimaging can open new avenues for probing social decision-making and social trust in the troubled minds (Razin & Feigh, 2021; Rilling & Sanfey, 2011).

Ultimately, the concept of MIE suggests a shift in how we think about intelligent systems. Rather than viewing intelligence as the property of isolated agents optimizing static objectives, this framework emphasizes adaptive agents embedded in interactive environments, continuously learning and predicting one another across multiple levels of representation. As human and AI become increasingly intertwined, understanding these multilevel interaction dynamics may become a central problem for neuroscience, AI, and the science of complex systems.

8. Conclusion

In summary, our proposed pyramid framework reframes equilibrium as a multiscale phenomenon of interactive intelligence, linking neural computation, cognitive inference, and observable behavior across both biological and artificial agents. This perspective opens the door to a richer theory of game dynamics in NeuroAI, where equilibrium is understood as a stable configuration emerging from coupled processes across levels and agents. More broadly, the framework suggests that many human-AI systems should be designed not merely to optimize performance, but to reach stable interactive equilibria that humans can easily predict and trust.

CRediT authorship contribution statement

Zhe Sage Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Yuchen Shen:** Formal analysis, Investigation, Software, Visualization, Writing – original draft, Writing – review & editing; **Quanyan Zhu:** Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Human-LLM co-adaptation

As a simple illustration of human-LLM co-adaptation, let $x_t \in [0, 1]$ denote the specificity of the human prompt at time t, and let $y_t \in [0, 1]$ denote the degree to which the LLM's internal representation is aligned with the user's intended task.

Let's assume that the interaction utility is defined as

$$U_t = 1 - (x_t - y_t)^2,$$

so that task performance is maximized when prompt specificity and model alignment match (Fig. 2A). The human updates prompt specificity in response to task performance:

$$x_{t+1} = x_t + \alpha_H \frac{\partial U_t}{\partial x_t} = x_t - 2\alpha_H(x_t - y_t),$$

where $\alpha_H > 0$ denotes the human adaptation rate.

In parallel, the LLM updates its inferred alignment state using the observed prompt:

$$y_{t+1} = y_t + \alpha_M(x_t - y_t),$$

where $\alpha_M > 0$ denotes the model adaptation rate.

Defining the mismatch term

$$d_t = x_t - y_t,$$

the coupled human-LLM dynamics can be rewritten as a one-dimensional contraction mapping

$$d_{t+1} = (1 - 2\alpha_H - \alpha_M)d_t.$$

Thus, in light of the contraction mapping theorem (Hunter & Nachtegaele, 2020), the interaction converges to equilibrium (i.e., $x^* = y^*$) if

$$|1 - 2\alpha_H - \alpha_M| < 1.$$

Thus, stable communication emerges as a fixed point of human-LLM co-adaptation (Fig. 2B).

Appendix B. Two-agent trust game computer simulations

Trust game setting. We conduct computer simulations of 150-round two-agent trust games based on AI-AI interactions. Each agent starts with an initial investment amount. At each round t, both players (agent 1: Investor; agent 2: Trustee) observe a multiplier ρ_t (default setting: $\rho_t = 1.5$) and choose investment ratios $a_t^{(1)}$ and $a_t^{(2)}$, yielding payoffs $r_t^{(1)}$ and $r_t^{(2)}$, respectively (where superscript denotes the agent identity). In each round, each player's payoff r_t equals ρ_t times the opponent's investment ratio minus their own investment ratio.

$$\begin{aligned} r_t^{(1)} &= \rho_t a_t^{(2)} - a_t^{(1)} \\ r_t^{(2)} &= \rho_t a_t^{(1)} - a_t^{(2)} \\ a_t^{(1)}, a_t^{(2)} &\in \{0, 0.01, 0.02, \dots, 1\} \quad \text{look-up table} \end{aligned}$$

Game behavior simulation. Without the loss of generality, we assume that the Trustee (Agent 2) is the target of our interest, whereas the Investor (Agent 1) represents the master agent that has more access to the game knowledge or various trustee-like players (therefore, Agent 1 can be pretrained based on simulated self-play data or prior experiences). At each Monte Carlo simulation of 150-run trust game, we first initialize the Investor's first-round investment ratio by drawing from a normal distribution with mean of 0.5 and standard deviation of 0.25.

Next, we assume that the Trustee's continuous expected investment ratio $a_t^{(2)}$ at round t is given by

$$\mu_t^{(2)} = 0.5a_{t-1}^{(2)} + 0.25a_0^{(2)} + 0.25a_{t-1}^{(1)}, \quad a_{0,A}^{(2)} \sim \mathcal{U}(\frac{1}{8}, \frac{3}{8}), a_{0,B}^{(2)} \sim \mathcal{U}(\frac{5}{8}, \frac{7}{8})$$

followed by a discretization operation based on a bin size of 0.01; $\mathcal{U}(c_1, c_2)$ denotes the uniform distribution between c_1 and c_2. During pretraining, distinct attractors are induced by varying the Trustee's initial investment fraction.

RNN architecture. We use two gated RNNs to represent two agents. The gated recurrent unit (GRU) is one type of RNN with gated mechanism (Cho et al., 2014). In our model setting, each RNN consists of 5 input units, 26 hidden units, and three parallel output heads.

For notation simplicity, the superscript will be ignored in the following equations unless ambiguity arises. The network input $\mathbf{x}_t^{(i)}$ $(i = 1, 2)$ is a five-dimensional vector. For instance

$$\mathbf{x}_t^{(i)} = \begin{bmatrix} 2a_{t-1}^{(i)} - 1 \\ 2a_{t-1}^{(3-i)} - 1 \\ \tanh(R_{t-1}^{(i)}) \\ 2\bar{a}_{t-5:t-1}^{(3-i)} - 1 \\ \Delta a_{t-1}^{(3-i)} \end{bmatrix}$$

where $a_{t-1}^{(i)}$ denotes the i-th agent's own investment ratio in the previous round; $\bar{a}_{t-5:t-1}^{(3-i)}$ denotes the opponent's averaged investment ratio over the preceding five rounds; $\Delta a_{t-1}^{(3-i)}$ denotes the change in the opponent's investment ratio from the penultimate to the previous round; $R_{t-1}^{(i)}$ is the agent's own reward in the previous round. All input elements are within the range $[-1, 1]$.

The input is next passed through a recurrent layer of 26 fully gated units

$$\begin{aligned} \mathbf{z}_t &= \text{sigmoid}(\mathbf{W}_z\mathbf{x}_t + \mathbf{U}_z\mathbf{h}_{t-1} + \mathbf{b}_z) \quad \text{(update gate)} \\ \mathbf{r}_t &= \text{sigmoid}(\mathbf{W}_r\mathbf{x}_t + \mathbf{U}_r\mathbf{h}_{t-1} + \mathbf{b}_r) \quad \text{(reset gate)} \\ \tilde{\mathbf{h}}_t &= \tanh(\mathbf{W}_h\mathbf{x}_t + \mathbf{U}_h(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \quad \text{(candidate activation)} \\ \mathbf{h}_t &= (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \quad \text{(GRU output)} \end{aligned}$$

where $\mathbf{h}_0 = 0$; $\{\mathbf{W}_z, \mathbf{W}_r, \mathbf{W}_h\} \in \mathbb{R}^{26\times5}$, $\{\mathbf{U}_z, \mathbf{U}_r, \mathbf{U}_h\} \in \mathbb{R}^{26\times26}$; $\odot$ denotes the Hadamard product. At the output layer, the network consists of three parallel output heads:

- **Actor policy head**, which applies softmax to obtain probabilities over 101 actions, whose outputs are evenly discretized into 101 choices between 0 and 1: $\{0, 0.01, 0.02, \dots, 1\}$. We also introduce two contexts $c_t \in (c_A, c_B)$, each of which is associated with a different policy. This is used for testing non-stationary policy in the presence of context switch: $c_A \to c_B \to c_A$.

$$\begin{aligned} \mathbf{y}_t &= \mathbf{W}_{\pi,c_t}\mathbf{h}_t + \mathbf{b}_{\pi,c_t} \\ \pi_t(j) &= \text{softmax}(y_{t,j}) = \frac{\exp(\beta y_{t,j})}{\sum_{i=0}^{100}\exp(\beta y_{t,i})} \quad (j = 0, 1, \dots, 100) \end{aligned}$$

  where $\mathbf{y}_t \in \mathbb{R}^{101}$ denotes the logit output, $\mathbf{W}_{\pi,c_t} \in \mathbb{R}^{101\times26}$ and $\mathbf{b}_{\pi,c_t} \in \mathbb{R}^{101}$; β denotes the inverse temperature parameter that controls the steepness of the softmax function or the exploration-exploitation balance. In each round, the action is sampled from the 101 possible values according to the probability $\pi_t(j)$.
- **Critic value head**, where $V_t \in \mathbb{R}$ is a scalar estimating the value of the current internal state: $V_t = \mathbf{w}_v^\top\mathbf{h}_t + b_v$, where $\mathbf{w}_v \in \mathbb{R}^{26}, b_v \in \mathbb{R}$.
- **Opponent's behavior prediction head**, $\hat{a}_t = \text{sigmoid}(\mathbf{w}_{a,c_t}^\top\mathbf{h}_t + b_{a,c_t})$.

In MARL, all unknown model parameters $\{\mathbf{W}, \mathbf{b}, \beta\}$ from two actor-critic RNNs are updated sequentially during the repeated-run game. The Trustee's RNN is randomized, whereas The Investor's RNN is initialized with a pre-trained model based on self-play simulations. During policy switching, Investor's RNN has a context-dependent weights $\{\mathbf{W}_{\pi,c_t}, \mathbf{b}_{\pi,c_t}, \mathbf{w}_{a,c_t}, b_{a,c_t}\}$ for Actor policy head and Opponent's behavior prediction head. Trustee's RNN uses empirical threshold-dependent parameters $\{\mathbf{W}_{\pi,\hat{c}_t}, \mathbf{b}_{\pi,\hat{c}_t}, \mathbf{w}_{a,\hat{c}_t}, b_{a,\hat{c}_t}\}$ based on the opponent's last-round Investment ratio (i.e., $a_{t-1}^{(1)}$) and the estimated context $\hat{c}_t$ (i.e., context switch is unknown to Trustee).

Attractor configuration. In model simulations, we assume that two policy contexts of Investor agent are encoded by two distinct attractors in the RNN, which correspond to two 26-dimensional parameter vectors $\mathbf{q}_A \in \mathbb{R}^{26}, \mathbf{q}_B \in \mathbb{R}^{26}$, followed by a hyperbolic tangent function: $\mathbf{v}_A = \tanh(\mathbf{q}_A) \in [-1, 1]^{26}, \mathbf{v}_B = \tanh(\mathbf{q}_B) \in [-1, 1]^{26}$. We also assume that the GRU state $\mathbf{h}_t$ is linearly mixed with the selected attractor state to update the final GRU state

$$\mathbf{h}_t^{(1)} \leftarrow (1 - \alpha)\mathbf{h}_t^{(1)} + \alpha\mathbf{v}_{A/B}$$

where $\alpha = 0.125$ denotes the attractor strength.

RNN pretraining. We use a simulated self-play strategy to pretrain the Investor actor-critic RNN using the on-policy Monte Carlo method (Silver et al., 2017; Sutton & Barto, 2018). Specifically, we first generate a batch of game data of 300×150 runs. Moreover, we use the following joint reward function R_t

$$\begin{aligned} R_t &= 0.5R_t^{(1)} + 0.5R_t^{(2)} \quad \text{(joint reward)} \\ R_t^{(1)} &= R_{t,payoff}^{(1)} + 0.5R_{t,alliance}^{(1)} \\ R_{t,payoff}^{(1)} &= r_t^{(1)} \\ R_{t,alliance}^{(1)} &= 1 - |a_t^{(2)} - a_t^{(1)}| \end{aligned}$$

and the following loss function averaged across 150 runs: $\mathcal{L}(1) = \frac{1}{150}\sum_{t=1}^{150}\mathcal{L}_t^{(1)}$, which consists of three components related to the output heads:

$$\begin{aligned} \mathcal{L}_t^{(1)} &= \mathcal{L}_{t,actor}^{(1)} + \mathcal{L}_{t,pred}^{(1)} + 0.2\mathcal{L}_{t,value}^{(1)} \\ \mathcal{L}_{1,actor}^{(1)} &= -\log \pi_1^{(1)}(a_1^{(1)}|\mathbf{x}_1^{(1)}) \\ A_t^{(1)}(j) &= stopgrad(R_t(j) - V_t^{(1)}) \quad (j = 0, 1, \cdots, 100) \\ R_t(j) &= 0.25(\frac{j}{100} + a_t^{(2)}) + 0.5(1 - |\frac{j}{100} - a_t^{(2)}|) \\ \mathcal{L}_{t,actor}^{(1)} &= -\sum_{j=0}^{100}\pi_t^{(1)}(j)A_t^{(1)}(j) - stopgrad(V_t^{(1)}) \quad (t \geq 2) \\ \mathcal{L}_{t,pred}^{(1)} &= (\hat{a}_t^{(2)} - a_t^{(2)})^2 \\ \mathcal{L}_{t,value}^{(1)} &= (R_t - V_t^{(1)})^2 \end{aligned}$$

where *stopgrad* function acts as an identity operation during the forward pass of RNN, but blocks the flow of gradients during backpropagation.

The model parameters are updated with stochastic gradient descent based on an Adam optimizer (learning rate 1/75, default momentum) (Kingma et al., 2015). Optimization is performed after each 150-round episode, with each subsequent episode regenerated under the updated policy. Thus, optimization is iterative rather than a one-shot global optimization over a fixed dataset. No petraining is adopted for the Trustee RNN.

Two-agent actor-critic reinforcement learning. We use the similar reward and loss function for actor-critic RL based on the observed data of two agents. For each agent, the opponent's initial hidden state in the RNN is assumed to be uninformative (i.e., zeros).

$$\begin{aligned} R_t &= 0.5R_t^{(1)} + 0.5R_t^{(2)} \quad \text{(joint reward)} \\ R_t^{(i)} &= R_{t,payoff}^{(i)} + 0.5R_{t,alliance}^{(i)} \quad (i = 1, 2) \\ R_{t,payoff}^{(i)} &= r_t^{(i)} \\ R_{t,alliance}^{(i)} &= 1 - |a_t^{(3-i)} - a_t^{(i)}| \\ \mathcal{L}_t &= \mathcal{L}_{t,actor}^{(1)} + \mathcal{L}_{t,actor}^{(2)} + \mathcal{L}_{t,pred} + 0.2\mathcal{L}_{t,value} \quad \text{(joint loss)} \\ \mathcal{L}_{t,actor}^{(i)} &= -\sum_{j=0}^{100}\pi_t^{(i)}(j)A_t^{(i)}(j) - stopgrad(V_t^{(i)}) \\ \mathcal{L}_{t,pred} &= (\hat{a}_t^{(1)} - a_t^{(1)})^2 + (\hat{a}_t^{(2)} - a_t^{(2)})^2 \\ \mathcal{L}_{t,value} &= (R_t - V_t^{(1)})^2 + (R_t - V_t^{(2)})^2 \end{aligned}$$

When differentiating the Actor term of the loss function, discrete action sampling precludes direct differentiation with respect to the sample drawn from the softmax probability. Instead, we use a Rao-Blackwell gradient estimator (Liu et al., 2019) and stochastic gradient descent with zero momentum to update all learnable parameters.

Stress induction. It is well known that stress can profoundly reshape social interactions by altering trust, cooperative, and decision-making. Traditionally, stress is simply viewed as an emotional response or physiological perturbation; within the game-theoretical framework, stress is treated as a latent computational feature that reshapes the equilibrium governing social interactions (Chen, 2026). For simplicity, we assume that stress reduces the learning rate in the Trustee agent (i.e., Agent 2 only), namely

$$\begin{aligned} \eta^{(2)}(s) &= \eta^{(2)}(0)(1 - 0.6\ s) \\ \eta^{(1)} &= \text{constant} \end{aligned}$$

where $s \in [0, 1]$ denotes the relative stress level (the larger value of s implies being more stressful). For each stress level, we run Monte Carlo experiments for different initial conditions.

Visualization via PCA and CCA. From the population activity of the RNN, we conduct principal component analysis (PCA) and visualize the activity in the dominant two principal components (PCs) subspace. Additionally, we standardize within-round 26-dimensional population activity and adopt the canonical correlation analysis (CCA) to examine the shared subspace between two RNN (Jiang et al., 2026; Zhang et al., 2025). We further project the activity of ongoing population activity of two RNNs onto the dominant CCA subspace and plot the canonical correlation (CC).

Equilibrium criteria. We use the following criteria for determining the equilibrium across level. At behavioral level, we assess the investment ratios of two agents in MARL, if the change of these metrics are sufficiently small (e.g., threshold 0.01) for a few consecutive runs,

$$(a_t^{(1)} - a_t^{(2)}) \approx (a_{t-1}^{(1)} - a_{t-1}^{(2)})$$

then we deem it as a behavioral equilibrium. At the cognitive level, we use the prediction of opponent's behavior as the empirical criterion, when the prediction errors are sufficient small (e.g., threshold 0.03) for a few consecutive (8/10) runs for two agents, we deem it as a cognitive equilibrium. At the neural level, we have examined two criteria: one based on distance to the fixed points of RNN's 26-dimensional population activity (contexts c_A and c_B), the other based on the CC in the dominant CCA subspace. The distance threshold is 0.05, and the CC threshold is 0.999. Finally, MIE is determined based on simultaneous satisfaction of three-level (behavioral, cognitive, and neural) equilibrium criteria.

The software for agentic trust-game simulations and video demonstration is available under the MIT License (https://github.com/shenyuchen2121-star/MIE_simulation_animation_and_code).

References

Akata, E., Schulz, L., Coda-Forno, J., Oh, S. J., Bethge, M., & Schulz, E., (2025). Playing repeated games with large language models. *Nature Human Behaviour*, *9*, 1380–1390.

Barak, D., & Costa-Gomes, M. (2025). Humans expect rationality and cooperation from LLM opponents in strategic games. arXiv preprint, https://arxiv.org/abs/2505.11011.

Camerer, C. F. (2003). Behavioral game theory. Princeton University Press.

Camerer, C. F. (2009). Behavioral game theory and neural basis of strategic choice. In *Neuroeconomics: Decision making and the brain* (pp. 193–206).

Chasnov, B. J., Ratliff, L. J., & Burden, S. A. (2025). Human adaptation to adaptive machines converges to game-theoretic equilibria. *Scientific Reports*, *15*, 29364.

Chen, Q., Zhao, D., Liu, C., Yang, M., & Shi, Y. (2024). Autonomous vehicles in mixed-autonomy traffic: Game theoretic human-like decision making countermeasures. *Complex Engineering Systems*, *4*, 25.

Chen, Z. S. (2026). Multi-agent game theory for precision psychiatry. *Front Psychiatry*. *17*, 1963963.

Cheng, L., Li, M., Tan, C., Huang, P., Zhang, M., & Sun, R. (2026). Computational game-theoretic models for adaptive urban energy systems: A comprehensive review of algorithms, strategies, and engineering applications. *Archives of Computational Methods in Engineering*, *33*, 2037–2114.

Cheng, L., Zhang, M., Wang, K., Yuan, M., Liu, Z., Wang, J., Zhang, K., & Huang, P. (2025). Evolutionary smart contracts for virtual power plant trading: Integrating prospect theory and multi-stage negotiation in cross-regional energy markets. *International Journal of Electrical Power & Energy Systems*, *173*, 111453.

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Proc. 2014 conf. empirical methods in natural language processing (EMNLP)* (pp. 1724–1734).

Cullen, M., Davey, B., Friston, K. J., & Moran, R. J. (2018). Active inference in OpenAI gym: A paradigm for computational investigations into psychiatric illness. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging, 3*, 809–818.

Fisac, J. F., Bronstein, E., Stefansson, E., Sadigh, D., Sastry, S. S., & Dragan, A. D. (2019). Hierarchical game-theoretic planning for autonomous vehicles. In *Proc. 2019 international conference on robotics and automation (ICRA)* (pp. 9590–9595). IEEE Press. https://doi.org/10.1109/ICRA.2019.8794007

Foerster, J., Chen, R. Y., Al-Shedivat, M., Whiteson, S., Abbeel, P., & Mordatch, I. (2018). Learning with opponent-learning awareness. In *Proceedings of 17th international conference on autonomous agents and multiagent systems (AAMAS'18)* (pp. 122–130).

Fudenberg, D., & Levine, D. K. (1993). Self-confirming equilibrium. *Econometrica*, *61*(3), 523–545.

Glimcher , P. W., Fehr, E., (Eds.) (2014). Neuroeconomics decision making and the brain. Cambridge, MA: Academic Press.

Goeree, J. K., Holt, C. A., & Palfrey, T. R. (2016). Quantal response equilibrium: A stochastic theory of games. Princeton University Press.

Harsanyi, J. C. (1967–1968). Games with incomplete information played by Bayesian players I-III. *Management Science*, *14*, 159–182, 320–334, 486–502.

Hatcher, A. (2002). Algebraic topology. Cambridge University Press.

Hofbauer, J., & Sigmund, K. (1998). Evolutionary games and population dynamics. Cambridge University Press.

Holt, C. A., & Roth, A. E. (2026). The nash equilibrium: A perspective. *The Proceedings of the National Academy of Sciences USA*, *101*, 3999–4002.

Hunter, J. K., & Nachtegaele, B. (2020). Applied analysis. Singapore: World Scientific.

Ismail, M. S. (2024). Exploring the constraints on artificial general intelligence: A game-theoretic model of human vs machine interaction. *Mathematical Social Sciences*, *129*, 70–76.

Jiang, M., Gu, L., Ma, M., Li, Q., Kao, J. C., & Hong, W. (2026). Neural basis of cooperative behavior in biological and artificial intelligence systems. *Science*, *91*, eadw8151.

Kelso, J. A., Dumas, G., & Tognoli, E. (2013). Outline of a general theory of behavior and brain coordination. *Neural Networks*, *37*, 120–131.

Khalvati, K., Park, S. A., Mirbagheri, S., Philippe, R., Sestito, M., Dreher, J. C., & Rao, R. P. N. (2019). Modeling other minds: Bayesian inference explains human choices in group decision-making. *Science Advances*, *5*, eaax8783.

Kingma, D. P., & Ba, J.,(2015). Adam: A method for stochastic optimization. In *Proc. international conference on learning representations (ICLR)*.

Lee, J. Y., Lee, S., Mishra, A., Yan, X., McMahan, B., Gaisford, B., Kobashigawa, C., Qu, M., Xie, C., & Kao, J. C. (2025). Brain-computer interface control with artificial intelligence copilots. *Nature Machine Intelligence*, *7*, 1510–1523.

Li, T., Zhao, Y., & Zhu, Q., (2022). The role of information structures in game-theoretic multi-agent learning. *Annual Review of Control*, *53*, 296–314.

Littman, M. L. (1994). Markov games as a framework for multi-agent reinforcement learning. In *Proceedings of 11th int conf. machine learning* (pp. 157–163).

Liu, R., Regier, J., Tripuraneni, N., Jordan, M. I., & McAuliffe, J. (2019). Rao-blackwellized stochastic gradients for discrete distributions. In *Proceedings of the 36th international conference on machine learning (ICML)*. PMLR Long Beach, California. *97*.

Loula, R., & Monteiro, L. H. A. (2020). A game theory-based model for predicting depression due to frustration in competitive environments. *Computational and Mathematical Methods in Medicine*, *2020*, 3573267.

Lowe, R., Wu, Y. I., Tamar, A., & Harb, J., et al. (2017). Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in neural information processing systems 30*.

Madduri, M. M., Burden, S. A., & Orsborn, A. L. (2021). A game-theoretic model for co-adaptive brain-machine interfaces. In *Proceedings of 10th international IEEE/EMBS conference on neural engineering (NER)* (pp. 327–330).

Madduri, M. M., Yamagami, M., Li, S. J., Burckhardt, S., Burden, S. A., & Orsborn, A. L. (2024). Predicting and shaping human-machine interactions in closed-loop, co-adaptive neural interfaces. *BioRxiv preprint*. https://www.biorxiv.org/content/10.1101/2024.05.23.595598v1.

Masuda, N. (2008). Oscillatory dynamics in evolutionary games are suppressed by heterogeneous adaptation rates of players. arXiv preprint. https://arxiv.org/html/0803.1023v1.

Maynard Smith, J. (1982). Evolution and the theory of games. Cambridge University Press.

McKelvey, R. D., & Palfrey, T. R. (1995). Quantal response equilibria for normal form games. *Games and Economic Behavior*, *10*(1), 6–38.

Meulemans, A., Nasser, R., Wołczyk, M., Weis, M. A., Kobayashi, S., Richards, B., Lajoie, G., Steger, A., Hutter, M., Manyika, J., Saurous, R. A., Sacramento, J., & Agüera y Arcas, B., (2025). Embedded universal predictive intelligence: A coherent framework for multi-agent learning. arXiv preprint. https://arxiv.org/abs/2511.22226.

Montague, P. R., Dolan, R. J., Friston, K. J., & Dayan, P. (2012). Computational psychiatry. *Trends in Cognitive Sciences*, *16*, 72–80.

Nash, J. F. (1950). Equilibrium points in n-person games. *Proceedings of the National Academy of Sciences USA*, *36*(1), 48–49.

von Neumann, J., & Morgenstern, O. (1944). Theory of games and economic behavior. Princeton University Press.

In A.Oulasvirta, P. O.Kristensson, X.Bi, & A.Howes, (Eds) (2018). Computational interaction. Oxford University Press.

Oweiss, K. G., & Badreldin, I. S. (2015). Neuroplasticity subserving the operation of brain-machine interfaces. *Neurobiology of Disease*, *83*, 161–171.

Pangallo, M., Heinrich, T., & Farmer, J. D. (2019). Best reply structure and equilibrium convergence in generic games. *Science Advances*, *5*, eaat1328. https://doi.org/10.1126/sciadv.aat1328

Parr, T., Pezzulo, G., & Friston, K. J. (2022). Active inference: The free energy principle in mind, brain, and behavior. Cambridge, MA: MIT Press.

Razakatinana, M., Kolski, C., Mandiau, R., & Mahatody, T. (2020). Game theory-based human-assistant agent interaction model: Feasibility study for a complex task. In *Proc. 8th int. conf. human-agent interaction (HAI'20)* (pp. 187–195). https://doi.org/10.1145/3406499.3415071

Razin, Y. S., & Feigh, K. M. (2021). Committing to interdependence: Implications from game theory for human-robot trust.arXiv preprint https://arxiv.org/abs/2111.06939.

Rilling, J. K., & Sanfey, A. G. (2011). The neuroscience of social decision-making. *Annual Review of Psychology*, *62*, 23–48.

Sandholm, W. H. (2010). Population games and evolutionary dynamics. MIT Press.

Shoham, Y., Rob Powers, R., & Grenager, T. (2003). Multi-agent reinforcement learning: A critical survey. Unpublished report. http://jmvidal.cse.sc.edu/library/shoham03a.pdf.

Shu, K., Mehrizi, R. V., Li, S., Pirani, M., & Khajepour, A. (2023). Human inspired autonomous intersection handling using game theory. *IEEE Transactions on Intelligent Transportation Systems*, *24*, 11360–11371.

Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., Dieleman, S., Grewe, D., Nham, J., Kalchbrenner, N., Sutskever, I., Lillicrap, T., Leach, M., Kavukcuoglu, K., Graepel, T., & Hassabis, D. (2017). Mastering the game of go without human knowledge. *Nature*, *550*, 354–359 (2017).

Su, Q., Wang, H., Xia, Y., & Wang, L. (2026). A multi-agent reinforcement learning framework for exploring dominant strategies in iterated and evolutionary games. *Nature Communications*, *17*, 490.

Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction. (2nd ed.). The MIT Press.

Wang, K., Cheng, L., Yin, M., Zhang, K., Wang, R., Zhang, M., & Sun, R. (2025). Evolutionary game theory in energy storage systems: A systematic review of collaborative decision-making, operational strategies, and coordination mechanisms for renewable energy integration. *Sustainability*, *17*(16), 7400.

Yang, Y., Song, C., Li, H., & Zhao, J. (2026). A large language model-based game equilibrium selection approach for human-machine shared driving. *IEEE Transactions on Intelligent Transportation Systems*, *27*(7), 7650–7665. https://doi.org/10.1109/TITS.2026.3692621

Yoshida, W., Dolan, R. J., & Friston, K. J. (2008). Game theory of mind. *PLOS Computational Biology*, *4*, e1000254.(2008).

Yoshida, W., Seymour, B., Friston, K. J., & Dolan, R. J. (2010). Neural mechanisms of belief inference during cooperative games. *The Journal of Neuroscience*, *30*, 10744–10751.

Zhang, X., Phi, N., Li, Q., Gorzek, R., Zwingenberger, N., Huang, S., Zhou, J. L., Kingsbury, L., Raam, T., Wu, Y. E., Wei, E., Kao, J. C., & Hong, W. (2025). Inter-brain neural dynamics in biological and artificial intelligence systems. *Nature*, *645*, 991–1001.

Zhang, Y., Aziz-Alaoui, M. A., Bertelle, C., & Guan, J. (2014). Local nash equilibrium in social networks. *Scientific Reports*, *4*, 6224.

Zhu, Q. (2026). Neuronic nash equilibrium: An EEG data-driven game-theoretic framework for BCI-enabled multi-agent behaviors. *Automation*, *7*, 78.

Zhu, Q. (2025b). Reasoning and behavioral equilibria in LLM-nash games: From mindsets to actions.arXiv preprint. https://arxiv.org/abs/2507.08208

Zhu, Q. (2026). Generative-conjectural LLM equilibrium for agentic AI deception with applications to spearphishing. In J. S.Baras, S.Papavassiliou, E. E.Tsiropoulou, & M. O.Sayin, (Eds.), *Game theory and AI for security* (pp. 1–25). Springer (*vol. 16223*). GameSec 2025. Lecture Notes in Computer Science. Cham.